FLIGHT TRAINING INSTRUCTION

BASIC FIGHTER MANEUVERING
T-45 STRIKE

2025

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CHIEF OF NAVAL AIR TRAINING
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Subj: FLIGHT TRAINING INSTRUCTION, BASIC FIGHTER MANEUVERING, T-45 STRIKE

1. CNATRA P-1289 (Rev. 06-25) PAT, "Flight Training Instruction, Basic Fighter Maneuvering, T-45 Strike" is issued for information, standardization of instruction, and guidance to all flight instructors and student military aviators in the Naval Air Training Command.

2. This publication is an explanatory aid to the T-45 Advanced Strike Flight Training curriculum and shall be the authority for the execution of all flight procedures and maneuvers herein contained.

3. Recommendations for changes should be submitted via the electronic TCR form on the CNATRA Web site.

4. CNATRA P-1289 (Rev. 03-23) Rev 4 PAT is hereby cancelled and superseded.

S. V. LAMOURE
By direction

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FLIGHT TRAINING INSTRUCTION

FOR

BASIC FIGHTER MANEUVERING (1V1)

T-45

P-1289

HOW TO USE THIS FTI

This Flight Training Instruction (FTI) is your textbook for the Basic Fighter Maneuvering (BFM) stage of your Advanced Strike pilot training and is the source document for all procedures related to BFM. In addition, it includes suggested techniques for performing each maneuver and making corrections.

Use your FTI to prepare for and afterward to ground events and flights. This information will help you prepare effectively: know all the procedures in the assigned section(s), review the glossary, and be prepared to ask your instructor about anything that remains unclear. Then you can devote your attention to flying the T-45. After a flight, review the FTI materials to reinforce your understanding and to clarify any difficult maneuvers or procedures.

Note that this FTI also contains information on emergencies related to this stage. This section of the FTI amplifies but ***does not supplant*** the emergency procedures information contained in the T-45 NATOPS manual.

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INTERIM CHANGE SUMMARY

The following Changes have been previously incorporated in this manual:

CHANGE NUMBER	REMARKS/PURPOSE

The following interim Changes have been incorporated in this Change/Revision:

INTERIM CHANGE NUMBER	REMARKS/PURPOSE	ENTERED BY	DATE

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BASIC FIGHTER MANEUVERING INTRODUCTION

When the P-80 finally became operational in the mid-1940s, WWII was coming to a close, as was seemingly the age of aerial dog fighting. While these aircraft never took part during operations in Europe, they became an integral part of combat operations in Korea. In November 1950, a P-80 flown by LT Russell Brown shot down a Russian MiG-15 in the world's first decisive all-jet aerial battle, acting as a reminder that Basic Fighter Maneuvering (BFM) was still a very real possibility in combat operations. With the maturation of aerial warfare evolving into Beyond Visual Range (BVR) engagement options, the decades that followed Korea exploded with the development and implementation of long-range air-to-air missiles that could potentially eliminate the possibility of Within Visual Range (WVR) engagements. During the Vietnam conflict, however, aircrew often found themselves unable to employ their BVR missiles, forcing weapons employment into the WVR arena, ultimately relying on BFM skills to defeat their opponents. The days of Korea and Vietnam are long past, and we have achieved amazing successes with the accuracy and reliability of our current long-range missiles. As we, and our enemies, continue to improve our BVR capabilities, a combination of electronic attack (EA), theater Rules of Engagement (ROE), switchology failures, or a momentary lack of attention to prescribed air-to-air timelines may ultimately bring us face to face with the enemy in a 1v1 engagement.

1. Combat Lessons Learned — Despite operating in an era of all-aspect BVR missiles, history has continuously proven that the majority of air battles are fought and won in the visual arena. Even in the largest engagements or multi-plane scenarios, for that brief moment when the decision is made to engage an opponent, we are involved in 1v1 BFM. Strike-fighter aircrew ***must*** be proficient at 1v1 air combat to minimize time-to-kill and ensure they leave merges unscathed.

2. Develop Fundamental Tactical Skills — Through BFM we are allowed to practice briefing, debriefing, stick, rudder, and throttle mechanics, and tactical decision-making. The development of these core tactical skills and the confidence we gain in maneuvering our aircraft throughout its flight envelope improves our ability to perform and maintain situational awareness in other strike-fighter missions.

Fundamental tactics and maneuvers of air combat have changed little in the last 70 years. In this stage, we will introduce the classic fighter-versus-fighter maneuvers and discuss how to employ them in staged and dynamic situations. It is incumbent upon all strike-fighter aircrew to have a sound understanding of 1v1. The 1v1 BFM discussion will use a building-block approach, progressing from the basics behind maneuvering the aircraft, to weapons systems capabilities and limitations, and ultimately applying concepts specific to Perch and High-Aspect BFM.

By the time you complete BFM, you will not be an expert. That happens only in time through constant coaching, practice, and experience. BFM will probably be your most demanding stage of flight training, requiring immense concentration and attention to your instructors. You must go beyond mastering the procedures and concepts presented in the classroom; you must apply them in the air and begin to establish your skills as a strike-fighter pilot. BFM is in many ways an art form - the ultimate art form of aviation.

CHAPTER ONE
BFM ADMIN

100. INTRODUCTION

ADMIN encompasses the non-tactical portions of the flight. For standard ADMIN procedures, reference the SOP/Tac SOP and FORM FTI.

101. FINAL RENDEZVOUS

Recall the Rejoin procedures from the Tactical Formation stage below:

The Rejoin gets the Fighters back into Parade formation prior to returning to base. It will be executed from Combat Spread and initiated at the "fence-out" call from Lead. If in a position other than Combat Spread at the "fence out" call, utilize Tac Form corrections to establish Combat Spread position (sucked positions do not require corrections as join up may be executed from this sight picture). If beginning a tactical rejoin from an acute position, avoid closure into Lead until on or aft of Lead's 3/9-line. Expect Lead to be at 80% N2 as you execute the rejoin, conditions permitting. It is critical to maintain sight of Lead at all times during the join. ***Within a half mile, never go 'belly up' for safety of flight.***

The ultimate goal of the Rejoin is to safely maneuver your aircraft to a position in which you can execute a join you have been taught previously. To execute the rejoin from Combat Spread, turn to establish a cut into Lead and descend. Modulate throttle position as appropriate for the unknown airspeed rendezvous. As nose to tail separation is established, maneuver the aircraft to a position in which a running rendezvous or CV rendezvous can be executed. Rejoins from sucked positions require a small cut into Lead due to nose to tail separation already established. During these maneuvers, be co-altitude to slightly below Lead to avoid going 'belly up' inside of a half mile and losing sight of Lead as you turn to parallel Lead's heading. ***Sight of Lead must be maintained.***

If Wing fails to execute the basics during the running rendezvous, an overrun may occur. An overrun occurs when the wingman flies past bearing line, ending up acute. If Wing is approaching Lead's bearing line too quickly, speed brakes may be required to capture bearing line. If unable to stop on bearing line, maintain approximately 250-300 ft abeam Lead while you decelerate, and work back to bearing line and once airspeed control is established, continue the join up. If in a CV rendezvous situation, execute the fundamentals (Altitude, Bearing and Closure), and use speed brakes if necessary to avoid an underrun.

Once Parade is established, the Rejoin is complete.

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CHAPTER TWO
BFM TAC ADMIN

200. INTRODUCTION

Tac Admin will be conducted the same as in the Tac Form stage.

201. TAC SOP

Be exceptionally familiar with the most current Tac SOP, it will contain all of the Tac Admin execution and comm standards required to be successful in this stage.

202. COMBAT CHECKLIST

All combat checks should be accomplished either on deck or en route to the area, short of selecting the MASTER ARM switch to ARM. In the T-45C, the MASTER ARM switch will be placed to ARM following the G-warm in order to provide the instructor an audible tone for when the trigger is pulled. Combat checks for the T-45C are as follows:

- Cockpit – Secure
- STORES page – A/A, GUN, RTGS boxed, WSPAN: 31 feet
- Master Arm switch – ARM
- Environmentals – Write down sun, winds, and cloud decks, and altimeter setting

203. PADS

Once established in the area, the section will use Combat Spread (CS) to maneuver in the operating area. The beginning of every BFM set will start with both aircraft meeting certain parameters. Figure 2-1 describes where Wing should be in relation to Lead. Expect to be at 300 KIAS until your Lead calls "Set XXX" or "Unload for knots."

	Position	**A**ltitude	**D**istance	**S**peed	Initiated From
SNAP SHOT DRILL	Abeam	Deck + 5k	1.0 nm	300	
FLATS	Abeam	Deck + 5k	0.5 nm	200	
ROLLER	Abeam	Deck + 6k	0.5 nm	200	
6K' SET	Abeam	Deck + 6k	1.5 nm	350	
9K' SET	Abeam	Deck + 6k	2.0 nm	400	Deck + 8.5k 300 KIAS
BUTTERFLY	Abeam	Deck + 6k Deck + 2k	1.0 nm	350	
ABEAM	Abeam	Deck + 6k Deck + 2k	1.5 nm	350	
All decks referred to for PADS will be the hard deck Allowable Deviations: Altitude +/-100 ft; Distance +/-0.1 nm; Speed +/- 10 kts; Bearing Line +/- 10°					

Figure 2-1 BFM PADS

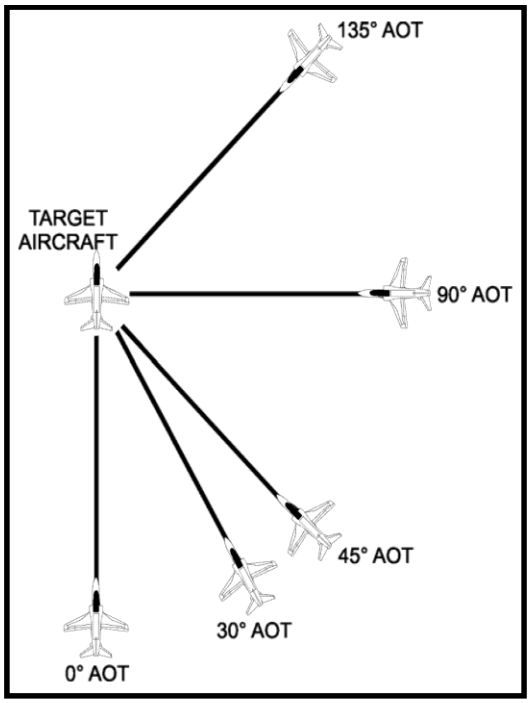

Figure 2-2 Angle Off Tail Overview

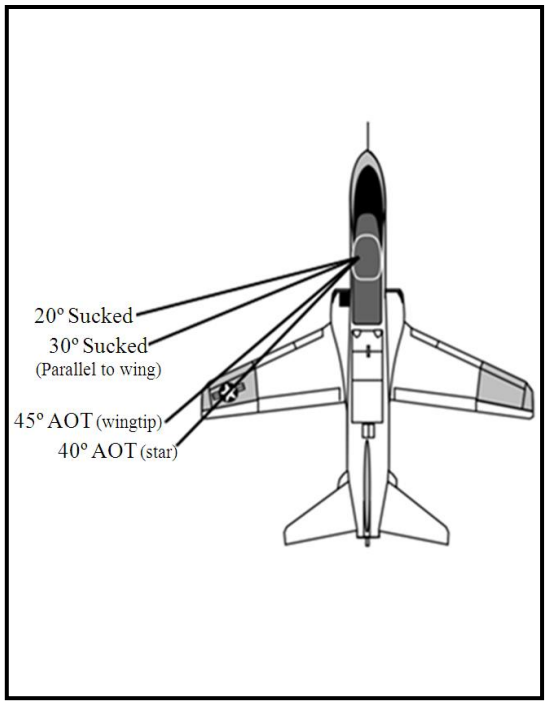

Figure 2-3 Angle Off Tail (Front Cockpit)

204. REFERENCE ANGLES

Use the above diagrams to reference Angle Off Tail (AOT), or degrees acute from Lead's 3/9-line when in level flight. Use the diagram below to reference AOT when in a defensive position.

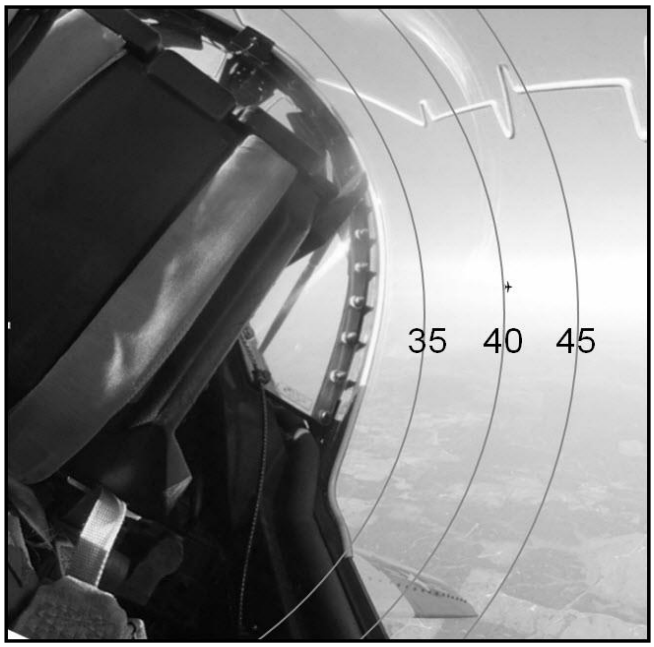

Figure 2-4 Angle Off Tail (Defensive)

Figure 2-5 40° Angle Off Tail

205. SPEED AND ANGELS

The Flight Lead will set the geometry for follow-on sets based on area management. The wingman's job is simple – maintain position. Oftentimes Lead will need to check the flight to a new heading (for geometry, area management, etc.). Utilize an energy-sustaining turn (if required) and get in position. If the set starts at 200 KIAS, maintain 300 KIAS until directed to slow by Lead; then, do whatever is necessary to stay in position as both aircraft decelerate.

Slight deviations from the PADS are acceptable, not to exceed the following:

- Altitude: ± 100 ft
- Distance: ± 0.1 nm
- Speed: ± 10 kts
- Bearing: ± 10°

The Lead will initiate the speed, and angels call when the wingman is in a good position. Echo the call when these parameters are met. If these parameters will not be met by the end of the transmission, call "stand by."

206. KNOCK-IT-OFF (KIO)

Upon the conclusion of each set, the trunk IP (on dual events) or the Flight Lead (on solo events) will call a ***Knock-it-Off***. Both aircraft will acknowledge the KIO (reference the Tac SOP for the proper comm). The priorities, in order, for the wingman following the KIO are:

1. Stand up the throttle (as appropriate), stop fighting, and maintain sight.

2. Hard turn, or pull as necessary, to the flow heading to maintain de-confliction and arrive in position.

3. **Capture the bearing line** and adjust nose attitude/throttle position to maintain position and match Lead in a 300 KIAS climb. Set abeam distance for next set when instructed to do so.

4. Climb to appropriate altitude.

5. With general SA, note G's and fuel while on heading and bearing line, in a 300 KIAS climb.

When in a climb, always reference 300 KIAS. For sets that start at slower airspeeds, Lead will advise when to decelerate.

207. VALID SHOT REQUIREMENTS

A fighter cannot just pull the trigger and consider the shot to be successful. Certain parameters must be met in order for the missile to track to the target or have bullets actually hit and inflict damage on the enemy aircraft. The requirements that must be met to consider it a valid shot are in the list below.

IR Missile:

1. Shooter within CNATRA Sidewinder Envelope.

2. Target in the HUD Field of View (FOV).

3. Wingman must not be in HUD FOV (shot de-confliction).

4. ***Pull the trigger (when steps 1, 2, and 3 met)***.

Gun Snap Envelope:

1. Shooter within snap gun envelope.

2. Pull trigger early (>1 sec. prior) to establish bullets downrange at target's distance.

3. Target must pass through pipper.

4. Two valid snap shots equal a kill.

Gun Tracking Envelope:

1. Shooter within tracking gun envelope.

2. Pull trigger with pipper on target.

3. One second of cumulative tracking time equals a kill.

Although not required for a valid kill, for training we must call our shots. The called shot must be appropriate for the weapon and envelope being employed after a valid shot has been taken:

1. IR missile shot – "Fox-2."

2. Snap gun – "Trigger down, snap (assessment), (missed high/missed low/looked good)."

3. Tracking gun – "Pipper's on, tracking…pipper's off."

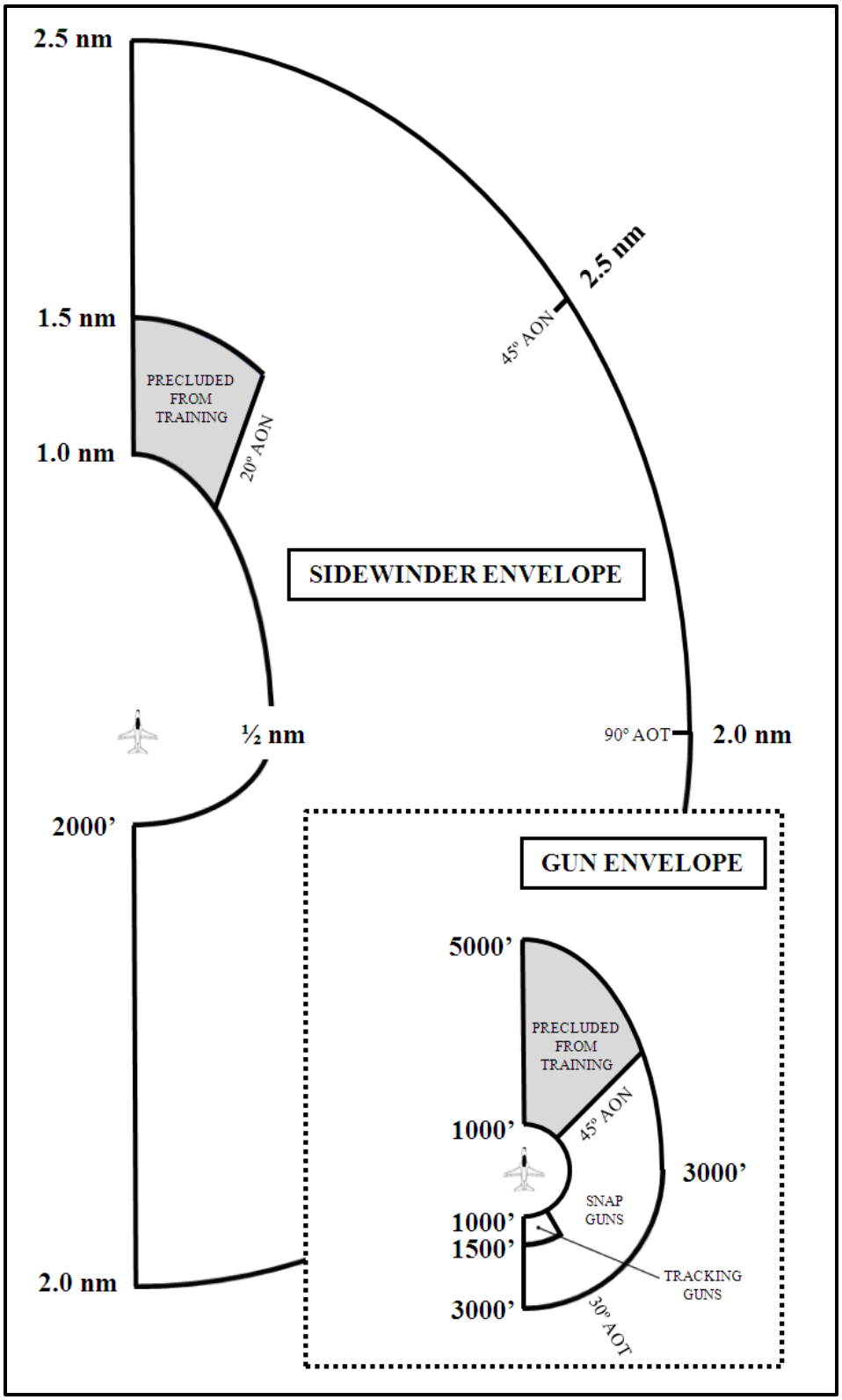

Figure 2-6 CNATRA Weapon Envelopes

208. DEFEATING SHOTS

In order to defeat shots certain requirements must be met.

Missile Defense (Missile D):

Simultaneously perform steps 1 and 2:

1. Call "Chaff, Flare" no sooner than two seconds prior to the adversary's sensor nose, and no later than the "Fox-2" call from adversary.

2. Place the lift vector on the adversary and perform a defensive break turn.

Guns Defense (Guns D):

Defeat shooter's Plane-of-Motion (POM) (Reference the defensive BFM Chapter for how to properly perform a "guns d").

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CHAPTER THREE
SAFETY AND EMERGENCY SITUATIONS

300. TRAINING RULES

The following Training Rules apply to all BFM and Section Engaged Maneuvering (SEM) training. They shall be strictly observed. These rules include those found in CNAFINST M-3710.7 and should be read prior to your BFM training. It is a requirement to brief training rules prior to each flight. Do not let this repetition lead to complacency. It is important to note that these rules were developed over a long period of time, and each is based not only on common sense, but also on situations where pilots were guilty of making serious and even tragic mistakes.

Administrative:

- Departure/spin and Compressor Stall/EGT/RPM. As the student you are responsible to brief OCF and engine stall emergency procedures.
  Scheduled face-to-face brief. Each experience in Air Combat Maneuvering (ACM) is unique, requiring all aspects of BFM and SEM flights to be briefed and debriefed thoroughly.
- ACM authorized by cognizant commander.
- Designated ACM area.

Currency: All In Flight Have Flown: As a student you will probably fall under the < 750 hr category.

- < 750 hrs FPT in type/class:
  - Once within previous 6 days.
  - Twice within previous 14 days (1 dynamic in Type/Model).
- > or = 750 hrs FPT in type/class:
  - Once within previous 14 days.
  - Twice within previous 30 days (1 dynamic in Type/Model).

Weather, Decks and Blocks:

Weather:

- Daylight (from 30 minutes past sunrise until 30 minutes prior to sunset), VMC, 5 miles visibility and a defined horizon.
- Cloud separation – 2,000 ft vertically and 1 mile horizontally.

Decks (brief MSL altitudes for working area):

- Hard Deck:
  - Minimum 10,000 ft AGL (or 5,000 ft above an under-cast). The under-cast shall be no higher than 7,000 ft AGL solo or 8,000 ft AGL dual.
- Soft Deck:
  - Minimum 5,000 ft above the hard deck.

- No slow-speed or high AOA maneuvering below the soft deck (less than 120 KIAS or more than 24 units AOA sustained for more than 3 seconds).

Blocks

- Established in assigned block by 10 nm without required SA on opposing force.

Comm Requirements

- Transmit, receive, monitor Guard, and ICS (multi-place aircraft).

Configuration changes other than speed-brakes are prohibited. You may not lower your flaps or gear.

Pre-commencement of ACM:

- Perform g-warm maneuver.
- Confirm:
  - Weather.
  - Announce local altimeter setting and any changes to decks or blocks.

Commencement of ACM:

Collision Avoidance:

- ***500 ft separation between all aircraft at all times***. This safety rule applies for training, both in the training command and in the fleet. In the real world, though, you must consider your adversary. For instance, if you maintain 500 ft on a head-on pass with an adversary who has forward-quarter weapons, you may be putting yourself directly into a weapons envelope AND giving away turning room. In the real world, know your adversary's capabilities.
- ***Always assume the other aircraft does not see you.*** You are personally responsible for collision avoidance at all times.
- Head-on pass:
  - Maintain the established trend; if no trend established, give way to the right to create a left-to-left pass.
- Broadcast your own intentions.
- Converging flight paths:
  - Nose-high goes high.
  - Nose-low has collision avoidance responsibility. Nose-low aircraft will ensure safe separation and must make way if the nose-high aircraft departs controlled flight, or somehow cannot stay nose-high (ballistic).
- Never intentionally maneuver to lose sight (blind lead turn). ***Do not make blind lead turns.*** A blind lead turn is when your nose is in front of the adversary's flight path, and you cannot see the adversary.
- Up-sun aircraft has the responsibility for collision avoidance. If down-sun aircraft loses sight, transmit "(call-sign) blind sun" and turn away from predicted collision bearing. If up-sun aircraft still has sight of the down-sun aircraft and safe separation can be maintained, the up-sun aircraft shall immediately broadcast "(call-sign) continue," otherwise Knock-it-Off. If you are in the sun, you are using a

tremendously powerful tactic because it blinds the adversary. But because the adversary is blind, it is your responsibility to maintain the safe separation. Also, if the weather is hazy, the sun creates a halo when you are looking down with the sun at your back. If the adversary is in the halo area, they cannot see you.

- If lost sight, transmit "(call-sign) blind" and turn away from predicted collision bearing. Other aircraft shall transmit "(call-sign) continue" or "(call-sign) blind (altitude)." If two aircraft have lost sight the first aircraft to transmit blind shall de-conflict via altitude.

  Knock-it-Off anytime de-confliction is not assured.

  - BFM Events Only: Knock-it-Off if both aircraft have lost sight.
  - SEM Events Only: If lost sight of adversary, use the term "no joy" vice "blind." Without a tally and visual on all fighters and adversaries, aircraft shall conduct belly checks at a minimum of every 90° of turn. Be sure to differentiate between "blind" and "no joy." "Blind" means you cannot see your wingman anywhere. It is a call made strictly to maintain safety. "No joy" means you can't see a threat aircraft described by your wingman.
- Call "ballistic" (for slow speed [<100 KIAS] reduced maneuverability).
- Brief CNATRA Weapons Envelopes or the following (whichever is more restrictive):
  - No head-on missile attacks inside 9,000 ft (1.5 nm) and 20° of the target's nose.
  - No forward-quarter gun attacks (within 45° of the target's nose).
  - Break off all gun attacks at 1,000 ft.

Terrain Avoidance:

- No guns defense (guns d) below the soft deck (aggressive nose-low, greater than 45° out-of-plane).
- Offensive aircraft will monitor the defensive aircraft's altitude, attitude, and airspeed and will break off the attack prior to pushing the defensive aircraft through the hard deck. Typically, a "watch the deck" call is sufficient to warn the other aircraft; this is for safety and to continue the fight.

<u>Termination of ACM</u>:

- ACM shall cease when:
  - Any training rule is violated.
  - "Knock-it-Off" is called by anyone, or an aircraft is rocking its wings.
- Knock-it-Off for:
  - Interloper.
  - Departure/spin. NATOPS calls for throttle to idle below 85 KIAS above 15,000 ft.
  - G-LOC (mandatory RTB).
  - Min altitude broken.
  - NORDO or ICS failure.
  - Overstress.

  - Bingo fuel. Don't forget about your fuel. You must keep your scan moving.
  - Inadvertent IMC.
  - Loss of situational awareness or any unsafe condition develops.
- 85 KIAS and decelerating for T-45C.
  - Training objectives attained. This is usually determined by the trunk IP (Lead if Solo).
  - Both aircraft lose sight
  - Approaching training area boundary.

Post Termination of ACM:

- Aircraft shall maneuver to maintain safety of flight and be aware of the high midair collision potential following the "Knock-it-Off" call.

301. CALLING THE PASS

Prior to every merge, both aircraft must know how to make a safe pass. In order for this to happen, each aircraft will acknowledge the pass. The most common calls are "high," "low," "right-to-right," and "left-to-left." Calling high or low should be fairly obvious (the wingman will echo the opposite call). A left-to-left call is, however, not as intuitive. The fighter that calls left-to-left, will be on the right side with the opposing fighter passing off the fighter's left wing (picture driving down the right side of the road with opposing traffic passing down the left side). All passes are Earth-stabilized (when inverted, the pass is reversed). Calling these passes needs to come early enough so both fighters have time to maneuver the aircraft safely. During a BFM engagement, the Lead will call a pass any time a trend is not clearly established. The student will respond and both aircraft will maneuver cooperatively to make a safe pass.

302. COMMON T-45C BFM EMERGENCIES

Due to the high load put on the T-45C during these engagements, there are times when certain aircraft systems will fail or underperform due to the stress. Here are some common emergencies:

OIL PRESS Warning:

This occurs typically following an overly aggressive unload to gain energy prior to an attack window entry. The negative G's experienced followed by a positive G onset will cause a momentary OIL PRESS Warning according to NATOPS. Knock-off the fight and execute the emergency procedures. It is acceptable to continue training after both fighters reset following a momentary illumination of the light after this kind of maneuver.

Compressor Stall:

The majority of compressor stalls in the T-45C happen during this stage of training. They are easy to prevent as long as aircrew are careful not to maneuver at high angles of attack, or maneuver above heavy buffet, when the engine is accelerating from low power settings, or when the engine is at high power settings. Typical characteristics of a stall are loud audible bangs,

chugs, or knocks from the engine. Knock-off the flight and execute the emergency procedures. The flight will RTB if either aircraft experiences this emergency.

CABIN ALT Warning:

Do not confuse the CABIN ALT warning light with an engine flameout or compressor stall. This is typically experienced under high AOA when engaged in the flats. Aircrew will hear the cockpit get quiet from the loss of cockpit ECS airflow, and there will be an audible warning tone and CABIN ALT warning light, with an associated loss of cabin pressure; RPM and EGT will, however, remain normal. Simply turning the ECS switch off for 1-2 seconds, then back on, typically clears the warning light and re-pressurizes the cockpit.

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CHAPTER FOUR
BASIC AERODYNAMIC REVIEW

400. INTRODUCTION

The measure of success in 1v1 air combat is simple, ***it is to kill or be killed***. In order to be successful, a fighter must maneuver into a position to employ weapons while denying the enemy the ability to employ weapons. The discussion will begin with the first part of the equation: maneuvering the aircraft.

401. LIFT LIMIT

The forces acting on an aircraft in flight are thrust, weight, lift, and drag (Figure 4-1). The interactions and changes between these forces define the motion of an aircraft through the air.

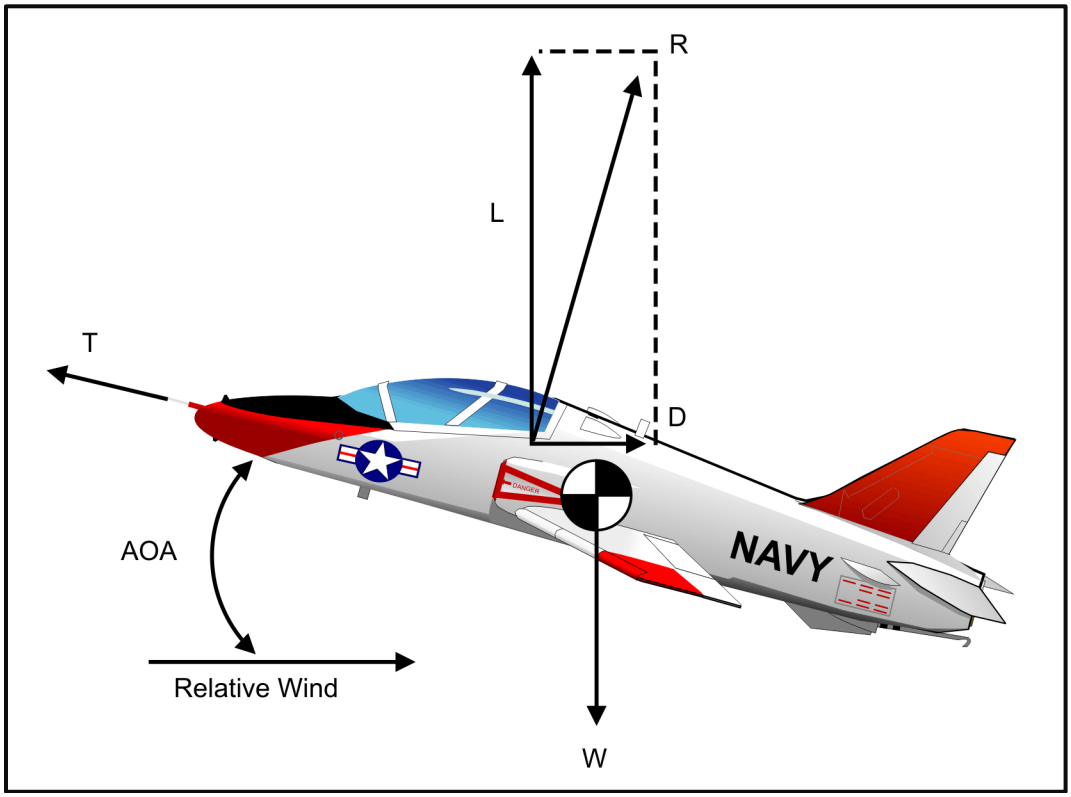


Figure 4-1 Aerodynamic Forces Acting Upon an Aircraft in Flight

During BFM, because thrust is usually at the maximum (MRT or full afterburner) and weight change is negligible at any given moment, thrust and weight will be considered constant during the aerodynamics review. This leaves the aerodynamic forces of lift and drag as the primary variables to consider when analyzing an aircraft's maneuvering performance.

Definitions for the discussion:

1. Lift - Created by the resulting pressure differential as air flows over a wing.

2. Coefficient of Lift (C_L) - A non-dimensional constant that is based on the shape of the wing. It is a function of Angle of Attack (AOA). Figure 4-2 depicts the relationship between C_L and AOA.

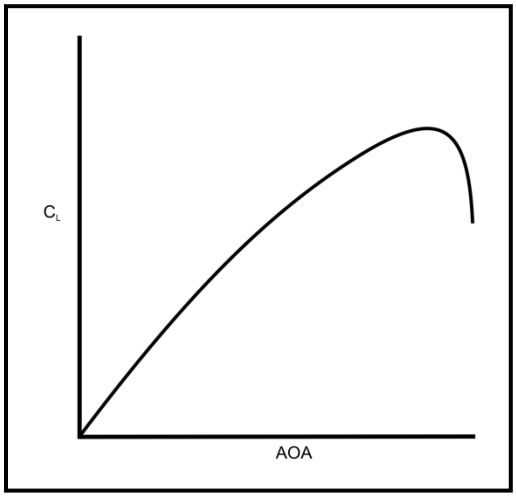


Figure 4-2 Angle of Attack

As AOA is increased, lift is also increased up to C_{LMAX}. The steep drop in C_L at high AOA indicates the point at which a portion of the wing stalls. C_{LMAX} occurs at 21 units AOA. ***At the lift limit the aircraft will be in heavy buffet.*** Lift and performance quickly diminish when pulling beyond C_{LMAX}.

402. ENERGY MANEUVERABILITY PERFORMANCE NUMBERS

Before discussing the maneuvers to use in order to achieve a positional advantage, aircrew must first be familiar with the performance of their own aircraft throughout the flight envelope. Two of the most important source documents for aircrew to reference when learning the full performance capabilities of the T-45C are Chapter 11 of NATOPS and the Energy-Maneuvering (E-M) diagram. Chapter 11 of NATOPS defines maneuvering characteristics and details the maneuvering limits that can prevent aircrew from departing controlled flight. The E-M diagram outlines aircraft turn performance during a level turn in the horizontal plane. It also allows aircrew to define tactical maneuvering options and reference airspeeds, as well as the structural and lift limit of our platform. Although engagements rarely remain level, understanding the principles behind the E-M diagram allows aircrew to make smart tactical decisions based on energy states.

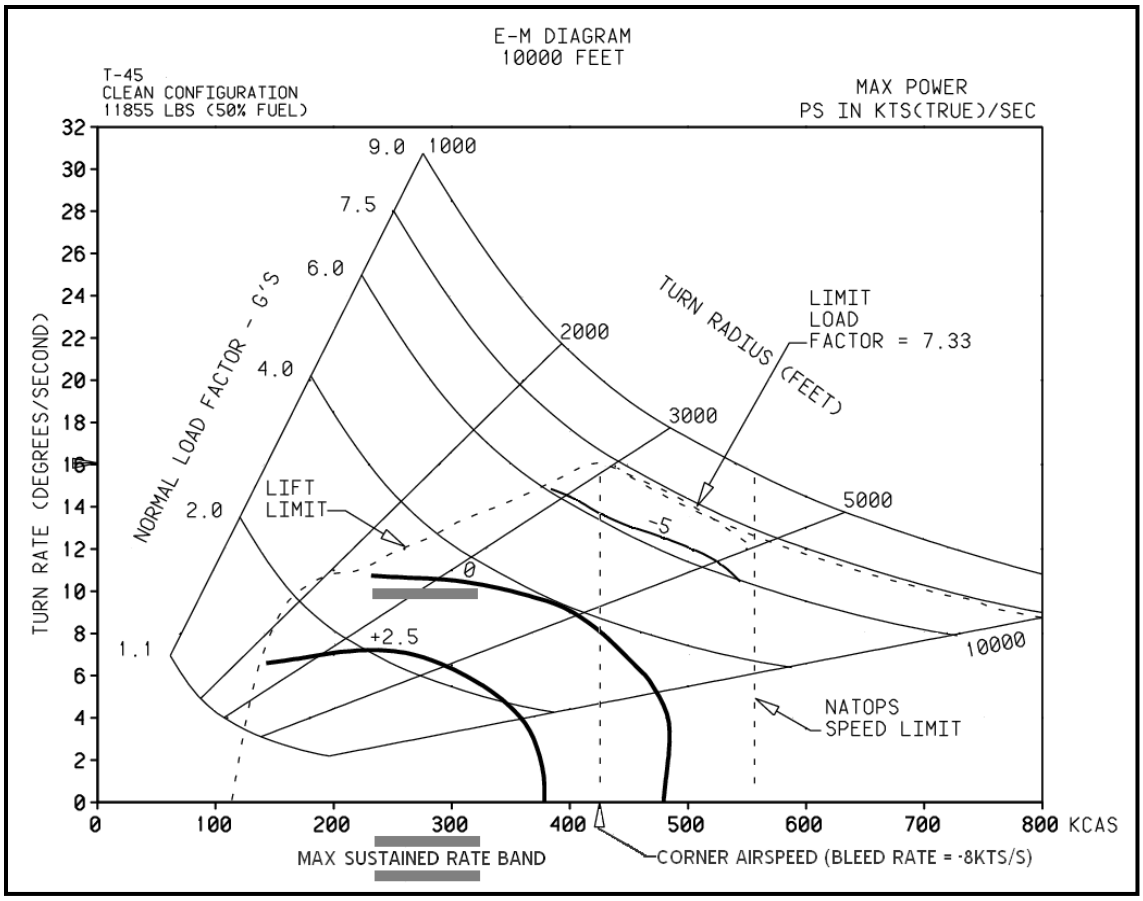


Figure 4-3 10,000 ft E-M Diagram

We are not going to build an E-M diagram in this chapter, but we are going to identify the main ideas and extract the most important information so as to arm you with the basic fundamental knowledge to execute BFM. Each diagram is specific for aircraft, configuration, and altitude, all of which can be found in the periphery of the diagram. Make sure to always read this information first to ensure you are deriving the correct performance characteristics for a specific aircraft in a specific scenario.

Although aircrew do not fly with an E-M diagram in the cockpit, it is imperative to commit certain performance numbers to memory. These numbers are required pieces of information. Do not show up at a merge without them!

403. T-45C PERFORMANCE NUMBERS

<u>Max Performance</u>:

- Pitch: Airspeed above corner airspeed: load limit pull, 7.33G (above 5,000 ft). Airspeed below corner airspeed: lift limit pull, 19-21 units AOA (rumble of buffet) pull.

- Roll: Unload while utilizing coordinated stick and rudder. The slower the speed, the more rudder deflection can be achieved due to lessoned aerodynamic loads. Due to this, the use of rudder at slower speeds becomes more beneficial in achieving maximum roll rates.

Acceleration:

Unload to 0G with MRT selected.

Corner Airspeed: 410 KIAS

This airspeed yields the aircraft's best instantaneous turn performance, but also sustains the highest bleed rates.

Corner Bleed Rate: 8 knots/second

This is the rate at which the aircraft will decelerate in a max performance pull at corner airspeed.

Compromise Performance:

Increased turn performance (better than energy sustaining) without the bleed rates associated with maximum performance.

- Below Corner Airspeed – Target 17 units AOA (light buffet).
- Above Corner Airspeed – Target 1 to 2 G less than maximum performance.

Best Sustained Turn Rate: 240-330 KIAS

This band is found where the Specific Power (Ps) curve = 0; i.e., the aircraft is sustaining energy. At the lower end of the rate band, the aircraft yields a small advantage in turn rate (0.5° higher at 240 KIAS compared to 330 KIAS). However, the aircraft's ability to trade airspeed for angles (energy excursion) is reduced below 300 KIAS. Furthermore, at slower airspeeds, the aircraft becomes more difficult to handle as it is more prone to stall and pitch buck. ***Initially targeting the upper portion of the rate band (300-330 KIAS), until an airspeed excursion is necessary, is prudent in most cases.***

Minimum Radius Airspeed Band: 140-180 KIAS (all altitudes)

Minimum Vertical Airspeed: 300 KIAS

Minimum vertical airspeed will vary based on altitude and proficiency. As a technique, targeting a light nibble of buffet pull will optimize the execution of a minimum airspeed vertical loop. In general, 300 KIAS is a good airspeed to target. Remember that aircraft and engine performance in an over-the-top maneuver is altitude dependent. Attempting to go over-the-top with less than minimum vertical airspeed may result in the aircraft going ballistic or even departing controlled flight.

4-4 BASIC AERODYNAMIC REVIEW

Tactical Vertical Airspeed: 350 KIAS

Tactical vertical airspeed corresponds to the airspeed required to initiate a tactical nose-high maneuver. Once the aircraft is nose high, aircrew still have tactical options available, including vertical extension, lift vector reorientation, or airspeed excursions.

404. VERTICAL MANEUVERING

E-M diagrams only depict an aircraft's turn performance in a level turn. Once the fight shifts into the vertical realm, we can no longer reference E-M diagrams alone to accurately represent the aircraft's turn performance, as the effects of gravity will affect turn rate, turn radius, and the bleed rates associated with a given maneuver.

Nose-High Maneuver:

Figure 4-4 illustrates the effects of gravity on vertical turn performance. The Load Factor (LF) is illustrated with a constant G force of 4 and is controlled by the pilot. Gravity (G) equals the effect of gravity on the aircraft. Finally, the Radial G (RG) is the combination of G and LF and denotes total performance.

Gravity can positively affect the turn rate and turn radius of an aircraft during a nose-high maneuver. The diagram shows a T-45C executing a pure nose-high loop. The pilot executing the maneuver maintains a constant 4G pull, but the aircraft overall performance (RG) must account for 1 unit of constant gravity. Remember the RG is the sum of the load factor the pilot applies and the one unit of gravity. Although the pilot pulls 4G at the beginning of the maneuver, the aircraft is fighting uphill against 1 unit of gravity and achieves only 3RG of turn performance. Conversely at the top of the maneuver, the aircraft achieves 5RG of aircraft performance, which produces a smaller turn radius and allows the aircraft to turn through more degrees per unit of time (greater turn rate). When the aircraft is bullseye nose-high, or nose-low, the effects of gravity are effectively zeroed, and a 4G pull results in 4RG of aircraft performance.

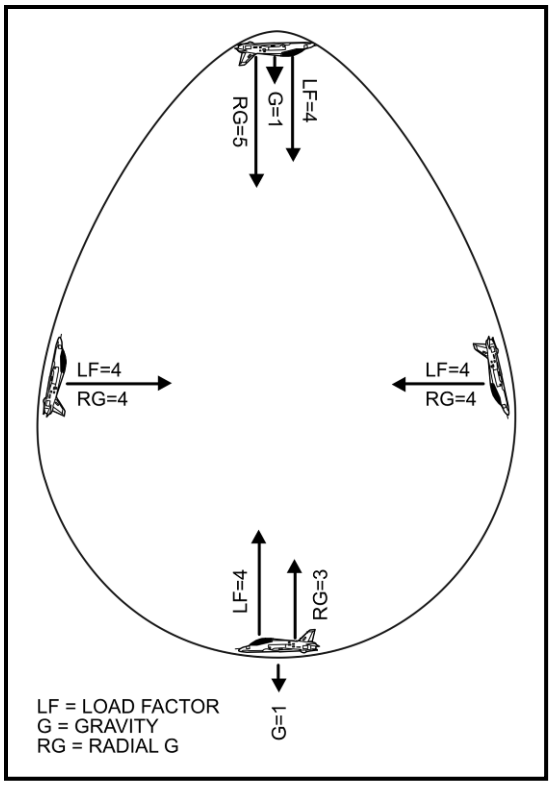


Figure 4-4 Vertical Maneuvering – The Egg

Nose-Low Maneuver:

While gravity can assist the aircraft's turn performance as it maneuvers nose-high, a combination of power, AOA control, and gravity can adversely affect turn performance as the aircraft begins to maneuver nose-low. The T-45C does not bleed energy in an extreme nose-low maneuver. The aircraft will accelerate when extreme nose-low despite max performing at MRT. Finally, an aggressive nose-low maneuver at airspeeds in excess of 350 KIAS will create a greater probability of the pilot succumbing to ALOC or GLOC without proper body positioning or Anti-G Straining Maneuver (AGSM).

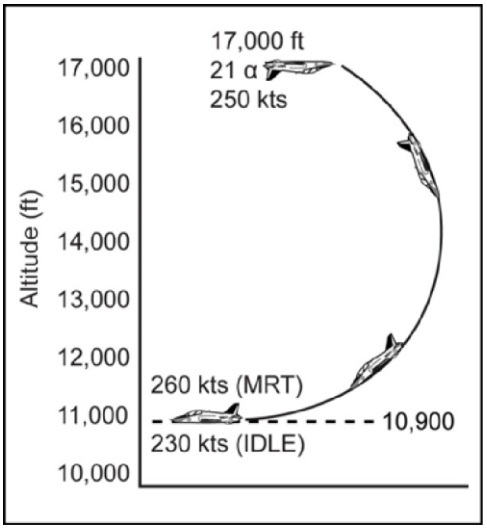


Figure 4-5 Nose-Low Maneuver (250 kts)

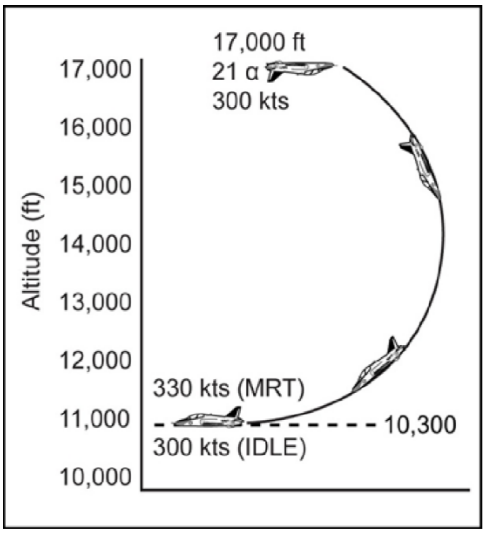


Figure 4-6 Nose-Low Maneuver (300 kts)

While performing a nose-low maneuver, the effects of gravity on turn rate and radius are reversed from the nose-high maneuver. If the aircraft maintains constant G, the turn rate will decrease, and the turn radius will increase throughout the maneuver. Figure 4-5 illustrates a T-45C executing a nose-low maneuver, max performing on the lift limit at MRT. Due to the change in radius, the aircraft's post will shift from its original position. This lateral separation between the posts needs to be accounted for when timing the nose-low maneuver, as is the case during the slow-speed merge discussion in High-Aspect Basic Fighter Maneuvers (HABFM). If the pilot executes a pull that is above or below the lift limit, the turn radius will grow even larger.

A T-45C requires approximately 6,000-6,500 ft of altitude to execute a pure nose-low maneuver (Split-S) when max performing at 19-21 units AOA. Slower airspeeds will generate a smaller turn radius and require less altitude to go pure nose-low. While a T-45C at 350-400 KIAS will require 7,000-7,500 ft for a Split-S, the same jet at 200-250 KIAS will require only 5,500-6,000 ft. It is important to know that throttle position (idle vs. MRT) does not affect turn radius (i.e., altitude loss) in a max performance pure nose-low maneuver—airspeed dictates turn radius and altitude required. Keeping the throttle at MRT will generate approximately 30-40 knots more at the bottom of the turn than keeping the throttle at idle. Throttle position on the way down will depend on what BFM problem you are trying to solve (discussed later in set specifics).

<u>Slow-Speed Vertical Reversal</u>:

Typically, slow-speed (less than 150 KIAS) vertical reversals are used during BFM when reversing from a nose-high to a nose-low pitch attitude in minimal time and turn radius. Maintain the throttle at MRT and ensure to keep the nose tracking past vertical (bullseye nose high) outside +/- 20° of pure vertical. It is paramount to keep a light nibble of buffet pull. If an excessive pull is used that leads to pitch buck, or the pull is insufficient to keep the nose tracking back to the horizon, a departure may occur. Although the jet is quite controllable during this parabolic maneuver at slow airspeeds, the aircraft may still depart. If the aircraft ever starts to

move in a direction that was not commanded, immediately neutralize the controls, reduce the throttle to idle, and follow NATOPS Out-of-Control-Flight (OCF) procedures. ***Remember, NATOPS states that the throttle needs to be at idle for altitudes above 15,000 ft MSL and airspeeds below 85 KIAS.***

Deck Transition Rule of Thumb (10% rule):

The deck transition rule of thumb provides a conservative flight path profile to transition from a nose-low attitude to level flight without flying through the hard deck. When to use the rule of thumb will be discussed later in the FTI. With practice and experience aircrew will be able to perform a more aggressive transition, but these recommended wickets provide a good starting point. Note that the max degrees nose-low translates to 10% of the altitude above the deck.

Nose-Low	Altitude Above the Deck
50°	5,000 ft
40°	4,000 ft
30°	3,000 ft
20°	2,000 ft
10°	1,000 ft

CHAPTER FIVE
OFFENSIVE BFM

500. OFFENSIVE BFM OBJECTIVES

The ultimate goal of offensive BFM is to kill the adversary as quickly as possible. If this primary goal is not achieved, ensure a positional advantage is maintained for follow-on weapons employment. Finally, if the positional advantage is lost, transition to high-aspect BFM. Simply put, the goals of offensive BFM are, in order:

1. Maintain the offensive position.

2. Employ follow-on weapons.

3. Transition to HABFM if neutralized.

NOTE

> HABFM will not be executed until introduced on the first HABFM event.

501. INTRO TO BFM CONCEPTS AND DEFINITIONS

In order to achieve a victory, a fighter must maneuver into a position to employ weapons while denying the enemy the ability to employ weapons. Combining offensive concepts with the cues in the cockpit will dictate the timing and mechanics to maneuver the aircraft and employ a weapon.

<u>3/9-Line</u>:

The 3/9-line is an imaginary plane extending from wingtip to wingtip, perpendicular to the flight path. Positional advantage is generally defined by fighter's position relative to the adversary's 3/9-line, or vice versa.

<u>Vertical 3/9-line</u>:

Vertical 3/9-line is similar to the 3/9-line but oriented in the vertical plane; it is not represented by the lift vector. The vertical 3/9-line is 180 degrees off orientation from the weight vector. In a rolling scissors the vertical 3/9-line defines the horizontal overshoot.

Turning Room:

Turning room is any separation that exists between two aircraft. Exclusive-use turning room is any turning room that is available to only one aircraft in the engagement.

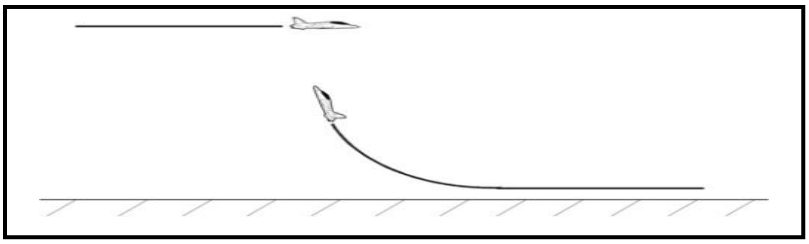


Figure 5-1 Exclusive-Use Turning Room

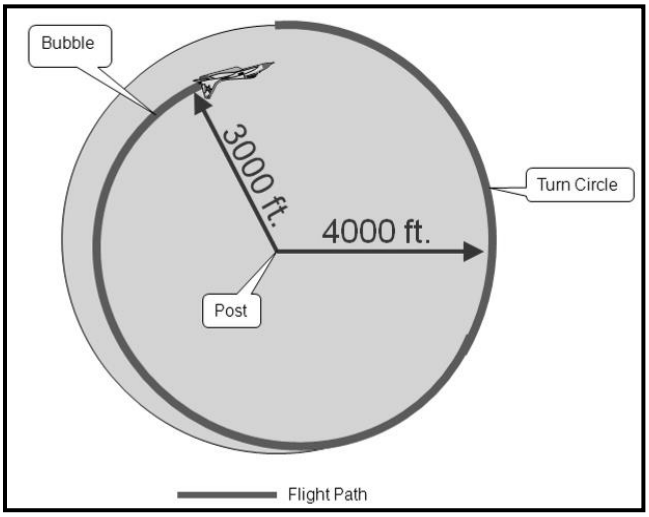


Figure 5-2 Turn Circle, Post, and Bubble

Turn Circle:

Turn circle is the actual turn an aircraft is scribing, based on current turn performance.

Bubble:

The actual turn an aircraft is scribing based on an instantaneous maximum performance turn given its current energy state. The visual cues in BFM are based on the aircraft's bubble. A template will be used during perch BFM briefs to represent the average bubble of the T-45C in a BFM engagement; approximately a 3,000 ft turn radius. ***An aircraft cannot turn inside its own bubble; this means that if a fighter is inside an adversary's bubble, the fighter is at least momentarily, immune from attack from the adversary. This immunity, and its impact on the current engagement, is important to note, both offensively and defensively, and is by far the most important BFM concept to understand.***

5-2 OFFENSIVE BFM

Post:

The post is the center point of the turn circle. The post does not define pursuit curves.

Plane-of-Motion:

Plane-of-Motion (POM) is the two-dimensional plane in which the turn circle resides. When an aircraft is at a high energy state, or executing a symmetrical pull in the vertical, its POM is a close approximation of the Plane of Symmetry (POS), or the plane in which the LV lies. When at lower energy states (i.e., 130 KIAS level, 60-degree AOB turn), POM may be more difficult to determine.

Control Zone:

Control Zone (CZ) is a cone-like area, 2,000 ft to 4,000 ft behind a maneuvering aircraft, 20° wide at the front and 40° wide on the back side, centered on the aircraft's flight path. If an attacking aircraft arrives in a defending aircraft's control zone with range, angles and closure under control, there is no single maneuver the defensive aircraft can execute that will deny the attacker the positional advantage.

Attack Window:

The Attack Window (AW) is a location inside the bubble where, if an attacker executes a smooth pull to stabilize the defender on the canopy, the attacker has the opportunity to arrive in the defender's CZ with Range, Angles, and Closure (RAC) under control. The pull to stabilize the defender on the canopy is known as Attack Window Entry (AWE) mechanics.

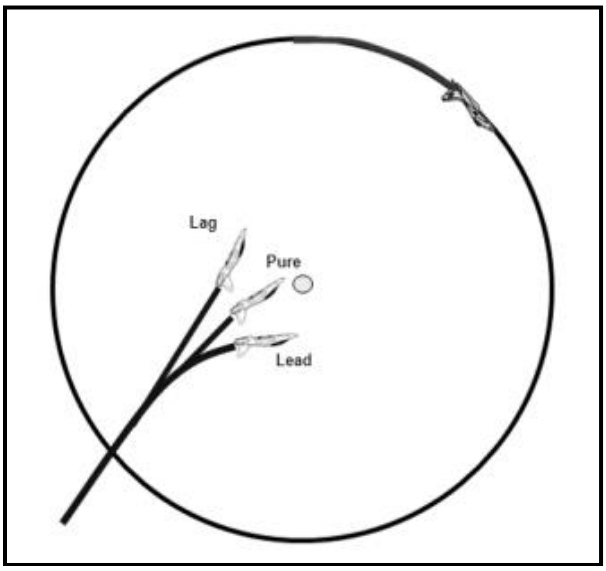


Figure 5-3 In-plane Pursuit Curves

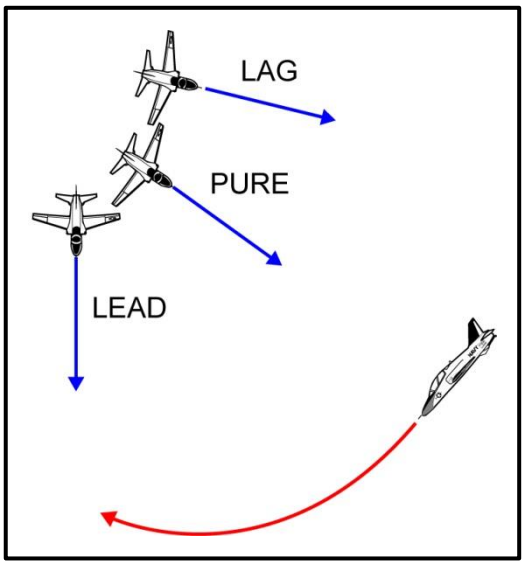
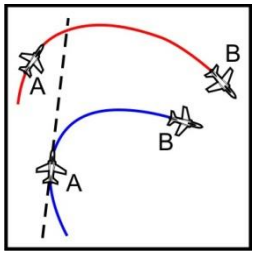


Figure 5-4 Out-of-plane Pursuit Curves

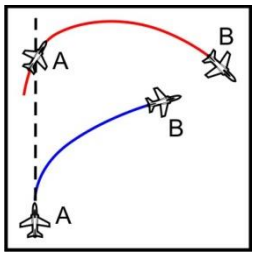
Pursuit Curves:

Pursuit curves are based on the fighter's nose position when in the adversary's POM, and the fighter's LV placement when not in the adversary's POM.



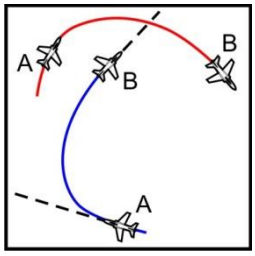
Lead Pursuit – Employ lead pursuit only for weapons employment, or when it is necessary to close nose-to-tail distance. Be ready to follow a lead pursuit maneuver with a lag pursuit maneuver to reduce V_c (rate of closure) and the AOT developed while flying lead pursuit.

- Nose or LV in front of the adversary.
- Used to rapidly collapse range and increase closure without regard to angles.
- Maintaining lead pursuit for too long can potentially lead to an in-close overshoot.



Pure Pursuit – Utilize pure pursuit (i.e., have the adversary in the HUD field of view) when you are ready to employ a weapon, or when possibly attempting to enter the adversary's bubble (aspect and range dependent).

- Nose or LV on the adversary.
- Used to employ weapons.
- Leads to a moderate collapse in range and increase in closure with little regard to angles.
- Maintaining pure pursuit for too long can potentially lead to an in-close overshoot.



Lag Pursuit – Mainly utilize lag pursuit maneuvers in order to correct back to the CZ, increase range, or manage closure.

- Nose or LV aft of the adversary.
- Used to manage the range, angle, and closure problem.
- Maintaining lag pursuit longer than necessary can give away offensive advantage.

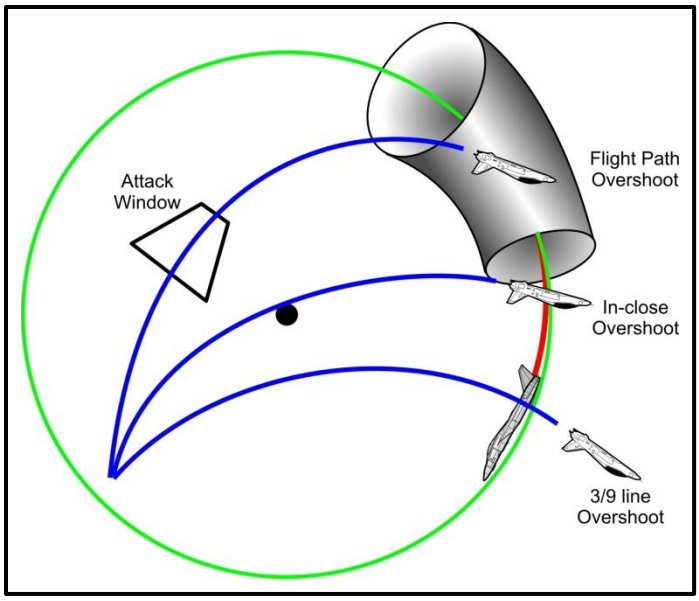
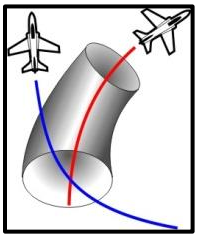
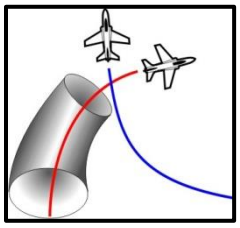


Figure 5-5 The Three Overshoots

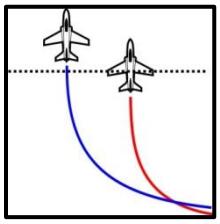
The Three Overshoots:



Flight Path Overshoot – Flying through the adversary's flight path. Ensure the flight path overshoot occurs within or aft of the confines of the CZ. This will deny a reversal opportunity.



In-close Overshoot – A flight path overshoot that occurs with high AOT inside the near portion of the defender's CZ, within 2,000 ft aft of the defender.



3/9-Line Overshoot – An overshoot of the defender's 3/9-line that results in a positional role reversal (i.e., passing from behind the adversary to in front of the adversary).

In-Close Flight Path Overshoots (ICFPOS) are extremely significant in that an instantaneous reversal by the defender may also cause a 3/9-line overshoot, possibly resulting in a role reversal. A flight path overshoot that occurs well aft of the defender is often insignificant because the defender cannot perform a role reversal.

If the overshoot occurs at the defender's CZ, the attacker will be able to maintain nose-to-tail separation by continuing the original turn to the defender's reversal point. By reversing at a CZ overshoot the defender actually helps the attacker solve some of the "degrees-to-go" problem.

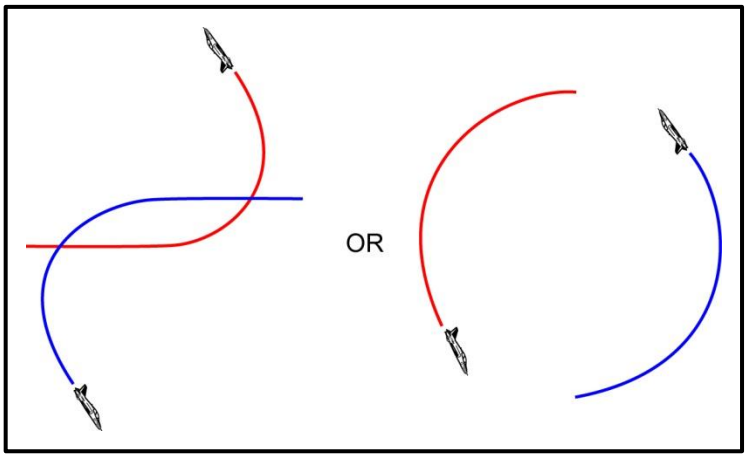


Figure 5-6 2C Flow

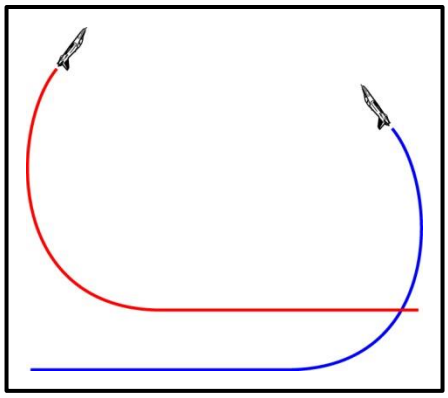


Figure 5-7 1C Flow

2C Flow:

While circle flow is often applied to neutral merges, the concept applies anytime two aircraft maneuver in relation to each other and the horizon. 2C flow is a rate fight and the angular advantage usually goes to the aircraft with the higher turn rate at its corner speed. A good technique often used is to trade some altitude for energy in order to continue maximizing turn rate. Often this kind of flow is referred to as a nose-to-tail, rate, or energy fight.

1C Flow:

During 1C flow, the fighter with the smaller turn radius will have the advantage. Aircraft will often pitch up, out-of-plane, to help minimize turn radius. This kind of flow is referred to as a nose-to-nose, radius, or positional fight.

Misaligned Turn Circles (MATC):

MATC generally applies to 2C flow and are created due to the fact that each aircraft's bubble is displaced over a different geographic point in space. Given the same exact turn performance, the geometry of MATC will enable an attacker's nose to eventually come to bear on the defender through pure geometry without having to perform an energy excursion.

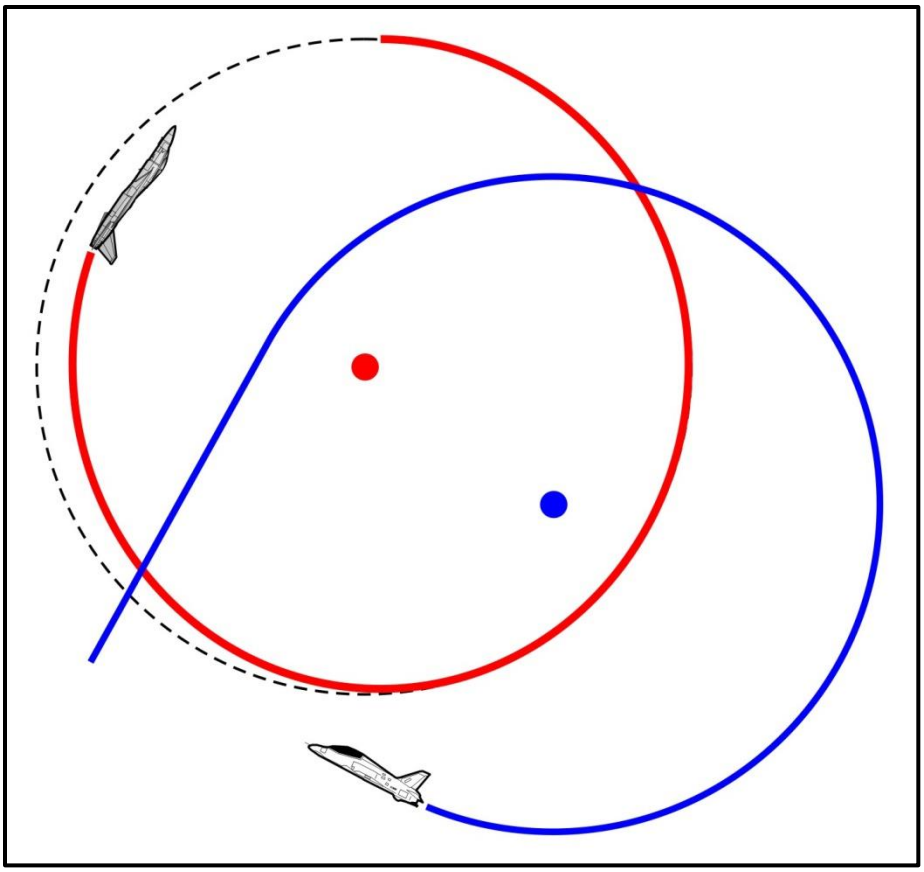


Figure 5-8 Misaligned Turn Circles

Energy Excursions:

An energy excursion is the conscious sacrifice of airspeed for angles. Fighters are always willing to make energy excursions in order to take or deny a shot, or to take or deny turning room.

BFM Cues:

Throughout an engagement, there are several visual cues that must be analyzed in order to make sound BFM decisions. The BFM problem can be measured by visual cues in the cockpit:

- Range: Being able to assess range is very important. This is accomplished by using the relative size of the aircraft or the information in the cockpit (A/A TACAN). Change in range will indicate closure or separation.

- Target Aspect (TA): The angular difference between the adversary's nose and your aircraft. An adversary that is pointing directly at your aircraft has zero target aspect. The ability to recognize the change in TA is very important.

- Angle off Nose (AON): AON is the angular measure that an aircraft is off your nose.

- Angle off Tail (AOT): AOT is the angular measure that an aircraft is off your tail. This is the best way to assess the angular problem of the engagement. AOT is used mostly in describing defensive BFM.

BFM Cue Assessment:

Being able to accurately assess the cues in a BFM engagement is very important. The most common cues are:

- Eyeball Calibration (EBC) – The most important aid that aircrew have in the cockpit is their own set of eyeballs. A visual assessment of range, AON, AOT, TA, and closure, as well as Attack Window Entry (AWE) and MATC, and bubble relationships are critical to success in BFM. The ultimate goal is to accurately recognize and assess all of the cues that are presented in a BFM engagement. This skill will develop over time.

- Aircraft Cueing Systems – The T-45C has few onboard sensors to assess range, TA, closure, etc. Utilize a visual sight picture, A/A TACAN, and the gun reticle to aid in range assessment.

Utilizing the Gun Reticle:

- Lead Angle Computing (LAC) – Used for tracking non-maneuvering targets at 1,000 ft range. The LAC reticle must be kept on the target for > 1 second. LAC will not be used during T-45C training.

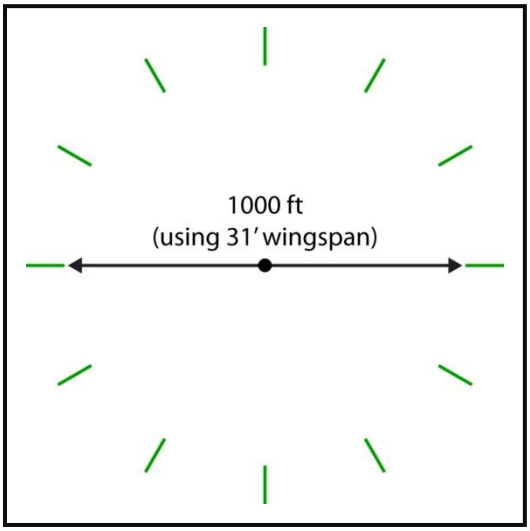


Figure 5-9 The Gun Reticle

- Real Time Gun Sight (RTGS) – Used for tracking a maneuvering target by aiming and keeping the reticle pipper ahead of the target (snapshot). The software positions the reticle on the point that a shell would reach in the time it takes to travel 1,000 ft.

502. SNAP SHOT DRILL (SSD) AS THE SHOOTER

The aerial gun was the first weapon employed in dog fighting and still remains the most difficult weapon to use in the BFM arena. The snap shot is used when attacking at medium to high angles off an adversary in an attempt to achieve a quick kill. The snap shot drill is a cooperative maneuver designed to teach employment of the gun at high AOT for the shooting aircraft and a timely guns d for the target aircraft. In order to achieve a valid snap, there are three problems the shooter must solve:

1. Plane of Motion
2. Range
3. Lead

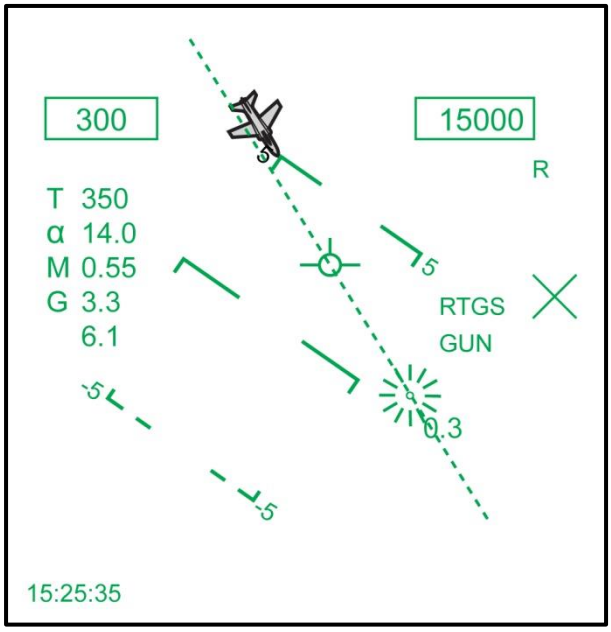


Figure 5-10 Snap Shot Drill (SSD)

Plane of Motion (POM):

Plane of Motion is the most important parameter to solve. It is typically the hardest to solve and therefore the easiest parameter for the target to defeat. The key to becoming accurate with the gun is to reference the defender's POM relative to the horizon. Achieve a good inside/outside scan, referencing the target's position (distance) above the horizon, relative to that of the pipper. Make large corrections for POM early by varying AOB to place the pipper the same distance above the horizon as the target. As range decreases, make smaller corrections. A good scan pattern is: target – horizon, horizon – pipper.

Range:

To solve range, tighten or ease the pull to have the target filling ¾ to all of the pipper, as "snap" is called (this indicates a 1,000 to 1,500 ft shot). If the shot looks like it will be long, ease the pull to travel further downrange. Then reset the pull. If the shot looks to be close, tighten the pull to achieve your gun solution earlier.

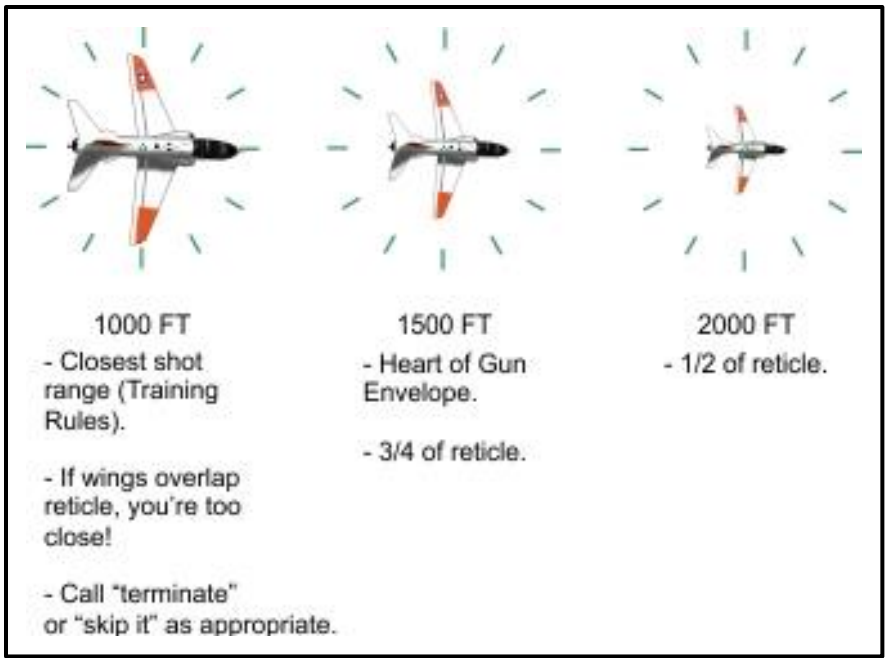


Figure 5-11 Estimating Range with the Gun Reticle

Lead:

Pulling the trigger when the target crosses the canopy bow and holding the trigger down until the target passes through the HUD is a good rule of thumb for establishing lead. This will account for ⅓ second pilot reaction time, ⅓ second gun spool-up time, ⅓ second bullet time of flight and allow the bullets to be at 1,000-1,500 ft downrange by the time the target crosses the pipper. If the trigger is pulled too early, bullets will be wasted. If the trigger is pulled too late, bullets will fall aft of the target. Therefore, in order to preserve ammunition, a disciplined and well-timed trigger squeeze is imperative. It is better to be a little early than late when pulling the trigger. Also, it is important to note that keeping the trigger down after the target passes through the HUD is also a waste of ammunition. Releasing the trigger and using the trigger finger to point at the target as you say "snap" is a good technique to save bullets.

Counter to a Guns Defense:

The shooter you will need to react quickly to counter an effective guns d. Decide early, then quickly roll to place the airspeed and altitude boxes two plane lengths out in front of the target's predicted flight path. The pipper will be unstable, and it may be difficult to determine a valid shot. With proper lead established, pull the trigger and evaluate the shot. Ideally the target would fly in between the airspeed and altitude boxes where the waterline symbol would be.

SSD Execution as the Shooter:

P – Abeam
A – Deck + 5,000 ft
D – 1.0 nm
S – 300 KIAS

The student will be the shooter on offensive BFM events. The typical flow for an offensive SSD will be four passes with a non-maneuvering target followed by two passes with a maneuvering target. Initially both aircraft will make their "speed and angels" call. Lead will then initiate the maneuver by calling "Ripper 11, in target." Wing will respond with "Ripper 12, in shooter."

The two aircraft will turn in towards each other utilizing a 2-3G pull. The shooter pulls to the inside of the turn while the target floats to the outside of the turn. The shooter controls the range while the target aircraft adjusts accordingly to maintain proper clock position for angles, targeting the 10 or 2 o'clock position. Once the geometry has been set, both the shooter and the target will maintain a smooth pull so that as the shooter's nose comes to bear, the target will allow the shooter to track forward to aft for the snap shot. The shooter will then call "trigger down…snap, (looked good / missed high / missed low)." The "trigger down..." call should be made the same time as the trigger squeeze. The "snap" call is made when the shooter assesses the bullets are at target range (e.g., as the target passes through the reticle) and is followed by the assessment of where the pipper actually was as it passed the target. It is important for a timely "snap" call so the target can make an appropriate evaluation of the shooter's nose position for future defensive maneuvers. Following the snap shot, both aircraft shall reverse their turns and adjust their pulls to complete what resembles a shackle turn. Both aircraft should strive to be back on the original PADS at the apex of the reversal. The shooter will then set the next pass.

On the maneuvering snap shots, Lead will call "Ripper 11, in target, maneuvering," followed by Wing calling "Ripper 12, in shooter, maneuver." During these passes, the target will attempt to maneuver out of plane in order to defeat the initial gunshot without subsequent maneuvers. The shooter will attempt to counter the defensive maneuver. After the gun defense and counter, both aircraft will maneuver cooperatively to achieve PADS for the next pass. In the case of a nose high or nose low defensive maneuver, both aircraft will maneuver cooperatively to put each other on the horizon prior to turning in for the next pass.

"Terminate" is a call used to temporarily cease training for administrative purposes. For example, "terminate" may be used to pause the SSD in order to maneuver the section away from an area border or redress the flight back to PADS. It shall not be used for any reason related to safety-of-flight. Following a "terminate," the flight lead will restart the SSD by simply calling, "Ripper 11, in target/shooter."

1. Lead: "Ripper 11 speed and angels right"
   Wing: "Ripper 12 speed and angels left"
   Lead: "Ripper 11 in target"
   Wing: "Ripper 12 in shooter"

2. Wing: "Trigger down, snap, looked good"

3. Lead: "Ripper 11 in target"
   Wing: "Ripper 12 in shooter"

4. Wing: "Trigger down, snap, looked good"

   (Shooter takes four shots, then…)

   Lead: "Ripper 11 in target maneuvering"
   Wing: "Ripper 12 in shooter maneuvering"

   Wing: "Trigger down, snap, missed low"

   (Shooter takes two more shots, then…)

   Lead: "Knock-it-off, knock-it-off"
   "Ripper 11 knock-it-off"
   Wing: "Ripper 12 knock-it-off"
   Lead: "Ripper flow XXX"
   Wing: "Ripper 12 XXX"

   Example terminate:
   Lead: "Ripper terminate"
   Wing: "Ripper 12 terminate"

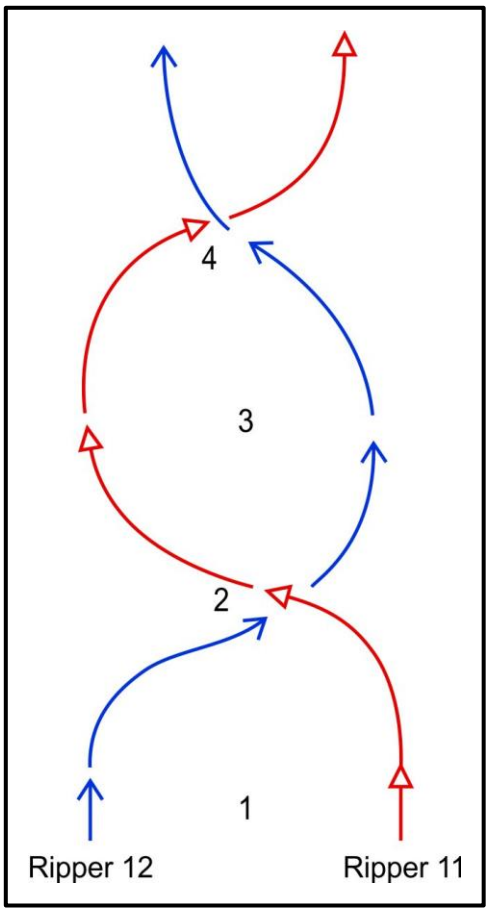


Figure 5-12 Snap Shot Drill

SSD Safety Considerations:

If it appears the shooter will employ inside 1,000 ft, but outside of the 500 ft bubble, either aircraft may call "skip it." The shooter will increase their pull, and the target will ease their pull. After a safe pass is affected, the shooter and target will maneuver cooperatively to set the next pass. If it appears the shooter will violate the 500 ft bubble, the first aircraft to recognize the impending safety of flight will call a "knock-it-off" and both fighters will immediately

de-conflict. The shooter will de-conflict by executing a loaded pull inside the turn, nose-low (in and down), max performing the aircraft. The target will de-conflict by executing a loaded pull away and go nose-high (up and away). The purpose of the knock-it-off is to ensure a safe pass and avoid a training rule violation.

503. OFFENSIVE FLAT SCISSORS (FLATS)

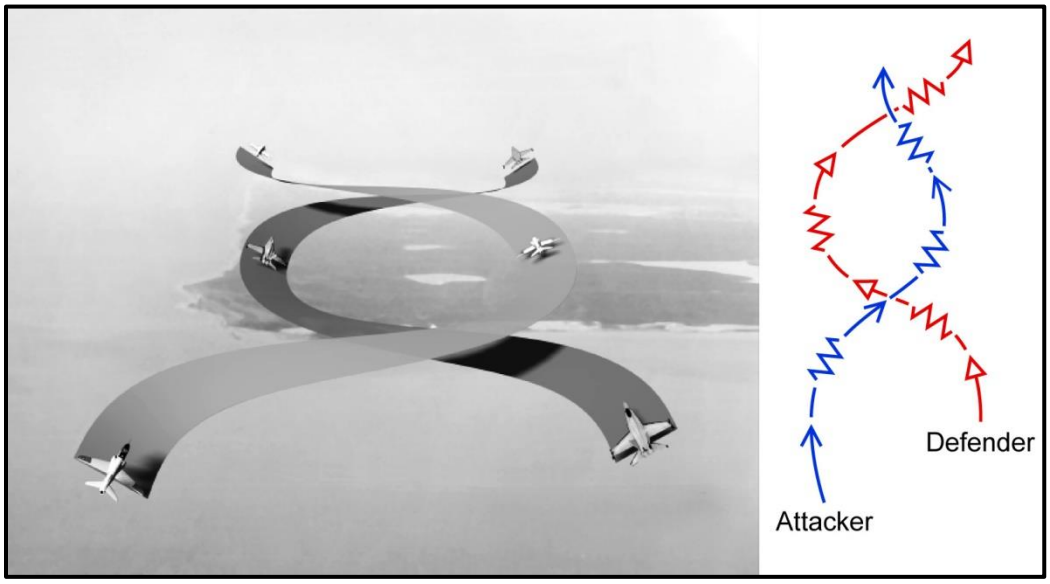


Figure 5-13 Flat Scissors Flow

The horizontal, or flat, scissors (flats) are a slow-speed, high-AOA fight where both fighters are in each other's bubble and are attempting to minimize their downrange travel to achieve a positional advantage. A flat scissors typically results from an in-close overshoot. There are several scenarios which may result in a flat scissors, such as a delayed or poorly performed lag maneuver, the offender pulling lead for too long on a guns tracking solution, poor BFM following a snap guns attempt, or follow-on flow from a rolling scissors. The flats may also be entered from a high-aspect merge where one fighter reverses to set 1C flow. The flats are not a 1C fight. In the flats both fighters are inside each other's bubble (the Adversary's target aspect and position on the canopy remain relatively constant) within roughly one half of one turn radius of each other with similar energy states and fuselages relatively aligned. In a 1C fight, the fighters are outside of each other's bubble which allows for the nose-to-nose type of fight. It is important to recognize the difference and fight using the appropriate mechanics.

Both aircraft will be attempting to decrease their downrange travel by performing a series of S turns. This effect of weaving in and out, or "scissoring," gives this fight its name. If both aircraft are flying at the same speed with one flying in a straight line and the other turning, the turning aircraft will eventually end up behind the aircraft flying straight because it has increased the distance it must travel before arriving downrange.

Kinetic energy (airspeed) is transferred to potential energy (altitude) by utilizing the vertical to aid in controlling both airspeed and downrange travel. Lift vector placement, typically placed aft of the adversary's CZ, airspeed and AOA control, and reversal timing are keys to success in scissor maneuvering. The fighter must utilize all three in the proper combination in order to prevail.

Offensive Flats Execution:

P – Abeam
A – Deck + 5,000 ft
D – 0.5 nm
S – 200 KIAS

Lead: "Ripper set 200"
"Ripper 11 speed and angels left"
Wing: "Ripper 12 speed and angels right"
Lead: "3, 2, 1, fight's on"

Lead: "Ripper 11 low"
Wing: "Ripper 12 high"

(Continue fighting until a logical conclusion when your IP calls for a KIO)

Flats Entry:

Lead will initiate the set by calling, "3, 2, 1 fight's on." On the "3," select MRT. On the "F" in lead's "fight's on," smoothly start a lift-limit pull. The adversary will delay the initial pull slightly (approximately two seconds after "fight's on") to establish a slightly offensive sight picture for the fighter. ***When both your pull and flight path de-confliction (in the form of altitude separation) are established, turn in towards the adversary***. This can be most efficiently accomplished at the fight's on by rolling to place the life vector above and aft of the adversary while pulling to the lift limit. Assuming the adversary gives an appropriate delay, altitude separation should be set. If the pull is weak or pitch buck is encountered, prioritize gaining altitude to ensure a 500' pass is made prior to turning toward the adversary. If no clear trend has been established, Lead will call a pass for the appropriate trend at the time, the wingman will respond, and the low aircraft (Lead) will give way to make that pass happen. The adversary will delay turning in until the wingman turns. While in the pull, trade airspeed for altitude while the nose tracks towards 40-50°nose high while turning into the adversary, thus minimizing turn radius and collapsing lateral separation. At the reversal (aft of the defender's 3/9-line), the fighter will no longer have the airspeed to continue the nose-high pull; when this occurs, the fighter has two options to recover to a sustainable nose-high position:

1. Overbank (roll slightly past 90° AOB) while maintaining the lift limit pull and get the nose tracking downward. Prior to 20° nose-high, roll upright to achieve 15-20° nose-high attitude at 130-150 KIAS.

2. Push forward on the stick to get the nose down, and then reset back stick to achieve 15-20° nose high at 130-150 KIAS.

A good rule of thumb is to initially pull 40-50° nose-high, and at 150-160 KIAS, begin the recovery maneuver. This first reversal should occur once aft of the defender's 3/9-line.

Flats Maneuvering vs. 1C Maneuvering:

While maneuvering in a true flat scissors the objective is to minimize downrange travel. While maneuvering in a 1C fight, the objective is to minimize turn radius. Simply flying as slow as possible wings level and having no energy for maneuverability is not the answer, nor is flying at 200 knots with hard turns level across the horizon. With minimum lateral separation and high fuselage alignment (flat scissors), flying slower than the minimum radius band will provide better results. This is due to the fight being a true high-AOA slow-speed flat scissors, vice a 1C fight (inside each other's bubble). With an increase in lateral separation however, the fighters need to increase airspeed into the minimum radius band to achieve optimum turn performance (outside each other's bubble). When flying on the faster side of the minimum radius band, the fighters will utilize more AOB while maintaining a lift-limit pull (slight nibble of buffet), to minimize turn radius and to ultimately threaten the adversary with the nose. This is especially useful when the fighter has an altitude advantage above the defensive aircraft, lateral separation exists, and the pull is away from the downrange direction of the fight. When maneuvering on the slower portion of the radius band the fighters will not be able to utilize as much AOB but will be minimizing downrange travel due to slower airspeed. This portion of the radius band is useful when fighter's noses are pointed downrange (with respect to the fight) and little lateral separation exists between both aircraft. Aircrew need to use a combination of both precise AOA control and smooth control inputs to fight their best jet.

Reversal Timing:

Execute reversals with the following criteria in mind:

1. To work in phase, turn aft of the adversary's 3/9-line prior to flight path crossing.

2. If a high track crossing rate (line-of-sight rate) exists, and the fighter is slower than the adversary, the fighter should turn a little early to deny lateral room and take advantage of the slower speed.

During reversals, utilize rudder inputs to turn towards the adversary and appropriate back stick to maintain optimum performance while monitoring attitude, airspeed, and AOA. Aircrew should be cautious to not over control the jet. If this occurs, one or both wings may stall, causing the nose to drop and the fighter to accelerate, flushing out downrange. If this happens, relax back stick pressure, level the wings, and re-establish the pull, working the nose up to get airspeed under control. Then reorient the lift vector as needed. Lift vector placement is more important the wider the fight becomes. As lateral separation decreases, precise airspeed control with an emphasis on generating the maximum amount of lift, to minimize downrange travel, becomes more of a priority. A good rule of thumb is that airspeed minus 100 is the max usable AOB (i.e., 130 KIAS ~ 30° AOB).

After maneuvering in the flats, the goal is to maintain the offensive position and gain weapons separation to employ a shot. The offensive positions from which to employ can be summarized in the following two situations with respect to energy and position.

Up and Aft:

Up and aft maneuvering is the most desirable. The offensive fighter has an energy advantage in the form of altitude and a positional advantage behind the adversary's 3/9-line. Look to work just slightly out of phase (not letting the adversary hide under the nose) and assess weapons separation. With sufficient separation, work your jet down to a valid weapons envelope using one of the following four methods:

1. Pulling beyond lift limit – This will increase drag and decrease lift, thus allowing the fighter to lose altitude without gaining airspeed. This is not desirable due to the T-45C being very difficult to control in this AOA regime, the increased possibility of departing controlled flight, or potentially causing an engine compressor stall.

2. Idle push over – Once weapons separation exists, select idle, extend the speed brakes, and push the stick aggressively forward, then reset back stick prior to the nose coming on the adversary and employ. Failure to reset back stick will result in excessive altitude loss and airspeed increase. Post shot, select MRT and retract the speed brakes, and execute a lift limit pull to get the nose tracking up and work back into the radius band. This technique is effective when desiring a quick shot; however, it is difficult to time and will need a good amount of weapons separation due to building airspeed and the risk of flushing downrange. ***Care must be taken when resetting the nose-high attitude and advancing the throttle in order to avoid an engine compressor stall.***

3. Reduce throttle and slowly work down utilizing precise AOA and airspeed control. This is most desirable; however, aircrew need to be patient and maintain precise control.

4. Overbank – With sufficient weapons separation, overbank the aircraft slightly to start the nose tracking down. The offensive fighter must visually project intercepting the defender's flight path with the nose to employ a weapon. It is important to maintain a lift-limit pull to avoid gaining extra airspeed and flushing out. After taking the shot, level the wings, and reset a nose-high attitude in order to minimize airspeed increase. Do not attempt a shot on the next pass. Work to regain lost weapon separation and employ again on a subsequent pass.

Down and Aft:

The offensive fighter should work in phase by performing early turns prior to flight path crossing. When the adversary reverses, the fighter should ease the pull momentarily to gain 10-15 knots, and then use this extra energy to work the nose up into a weapons envelope. This may take several passes to gain sufficient altitude and/or nose high attitude to employ weapon. This will require less AOB and potentially momentary pauses while wings level in order to gain airspeed to get the nose up. This will result in decreased nose to tail separation as the fighter works to gain a shot opportunity.

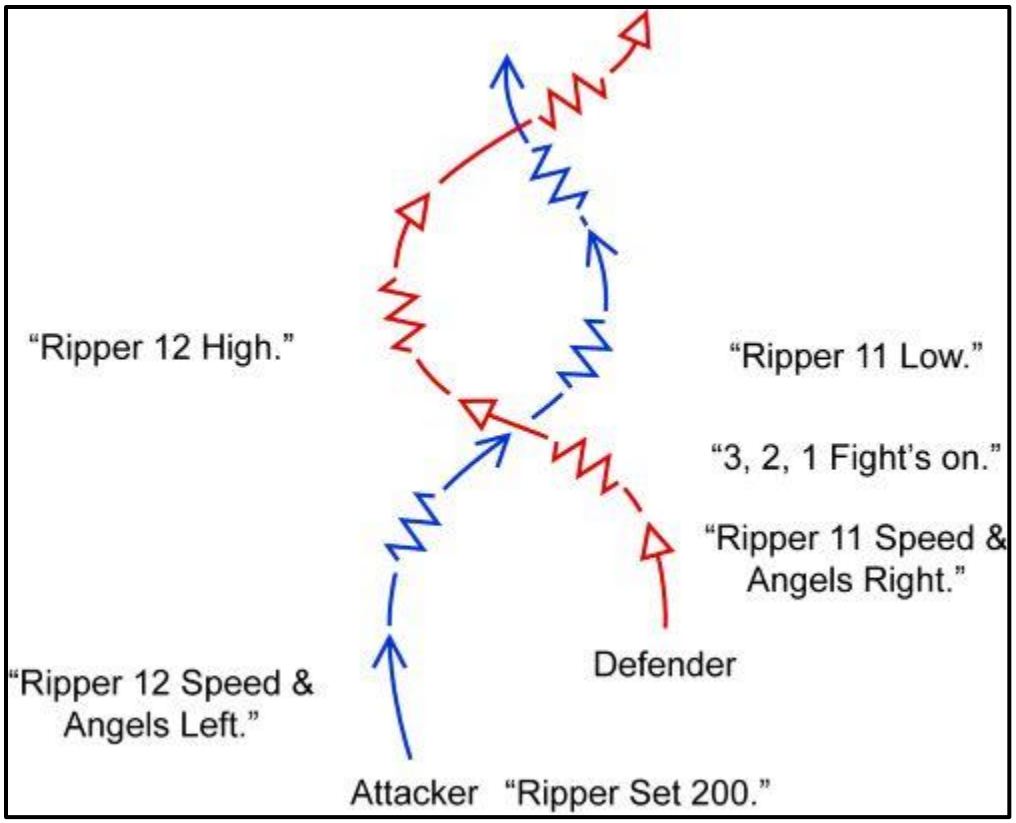


Figure 5-14 Offensive Flats Execution

Learning Objectives (emphasis) for the Flats:

1. PADS set up
2. 100% training rules adherence
3. AOA and airspeed control
4. Lift vector placement
5. Recognition of offensive or defensive position
6. Reversal timing execution
7. Shot opportunity recognition and mechanics
8. Guns d recognition and mechanics
9. Fight redefinition

504. OFFENSIVE ROLLING SCISSORS (ROLLER)

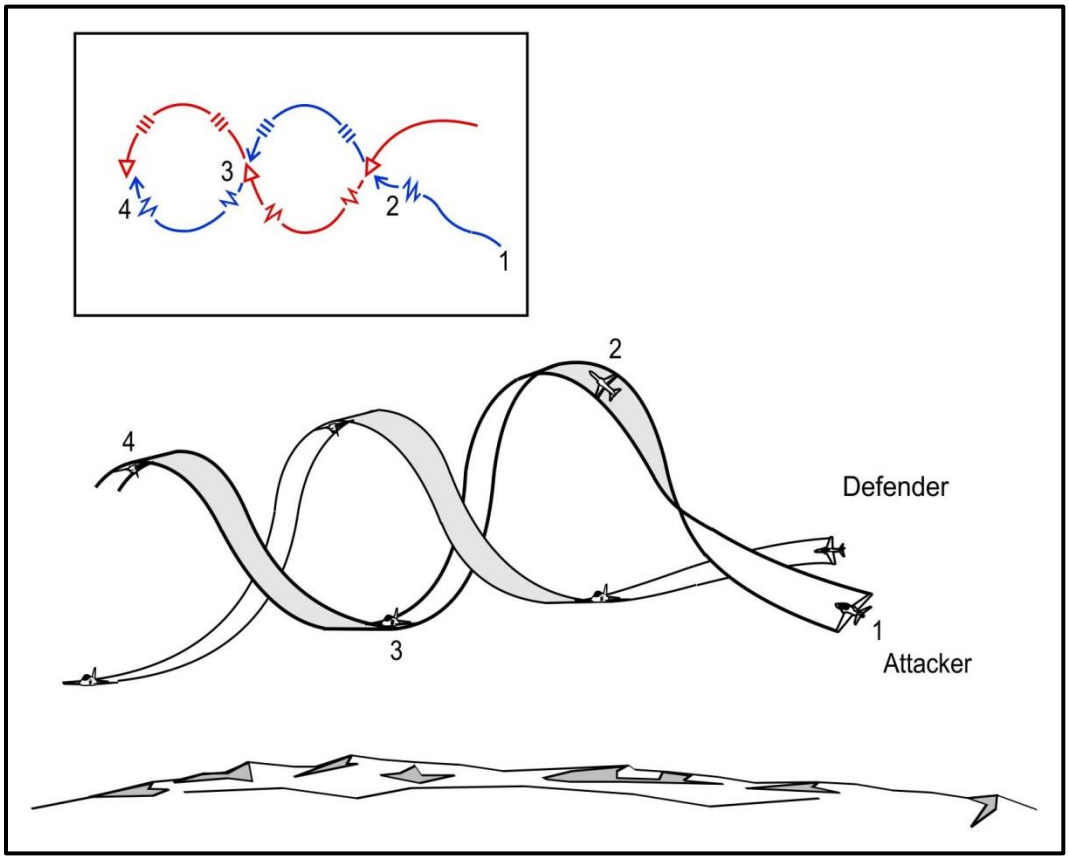


Figure 5-15 Roller Overview

The rolling scissors is a series of vertical and horizontal overshoots with both aircraft inside each other's bubble, where neither aircraft has sufficient energy to use the vertical. The rolling scissors may result from an in-close overshoot where the attacker attempts to stop downrange travel by pitching up into the vertical without sufficient energy to execute a pure loop. The resulting fight is a series of vertical and horizontal overshoots where both aircraft attempt to minimize both their down range and downhill travel while maximizing their turn rate and minimizing their turn radius in order to gain a positional advantage. Offensively, if the attacker finds themselves in a roller, they have made a mistake, which means they must strive to remain offensive or transition to high-aspect BFM. A rolling scissors is not a desirable maneuver for an attacker because it limits opportunities for weapons employment.

Roller Entry:

P – Abeam
A – Deck + 6,000 feet
D – 0.5 nm
S – 200 KIAS

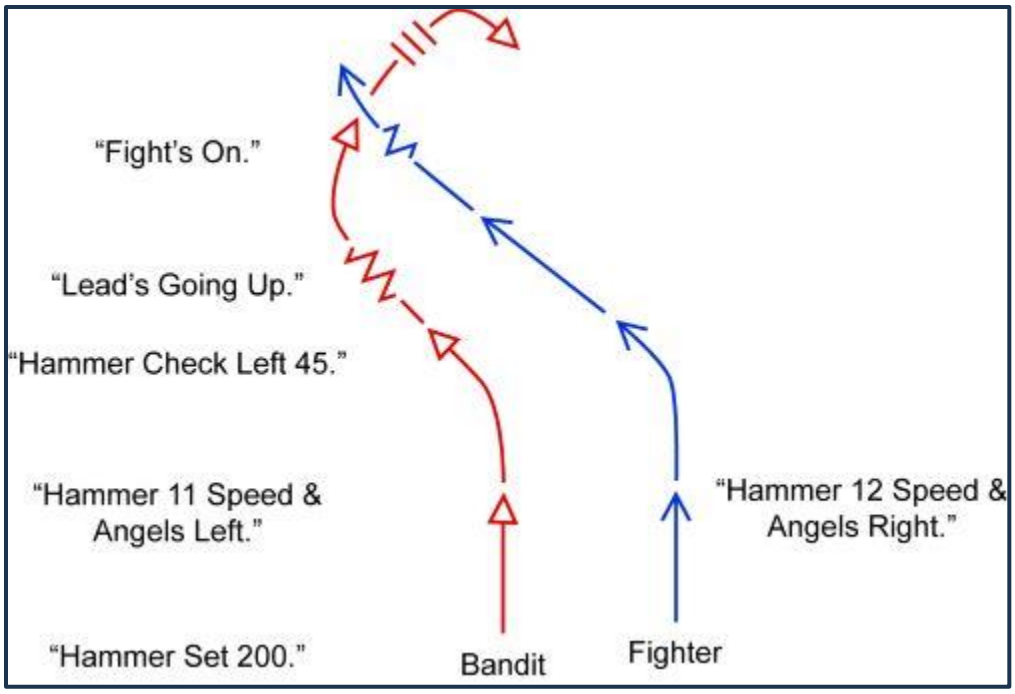


Figure 5-16 Rolling Scissors Entry

Lead will initiate the maneuver by calling "Hammer check (right/left) 45" to establish wing 45º AOT. After the aircraft arrive on heading, Lead will call, "lead's going up." At this time select MRT. Lead will maneuver to arrive over the top of the fighter in a defensive position and then call "fight's on."

Adversary: "Hammer set 200"
"Hammer 11 speed and angels left"
Fighter: "Hammer 12 speed and angels right"
Adversary: "Hammer check left 45"
"Lead's going up"
"Fight's on"

<u>Roller Maneuvering</u>:

When Lead calls "lead's going up," select MRT. At the "fight's on" call, perform a nibble of buffet wings-level pull to force a vertical overshoot. Approaching the vertical overshoot, roll to place the lift vector slightly above the horizon and max perform to force a horizontal overshoot. Be sure to keep the nose tracking above the horizon to preserve vertical turning room (altitude above the adversary). Prior to the horizontal overshoot, the fighter should note the adversary's nose position to determine whether they are offensive, neutral, or defensive:

- Offensive: If the fighter predicts the adversary will still be nose-low at the horizontal overshoot, then the fighter has an offensive advantage. Perform a corkscrew by reducing the throttle to idle, unloading, re-centering the stick to the most neutral position, leading with lateral stick and following through with bottom rudder to

quickly roll and place the lift vector on to slightly in front of the defender, and recapture a lift limit pull. A counter-correction may be required to stop rolling and set the lift vector. This will help the fighter gets in plane with the defender and reduce nose-to-tail separation. Keep the throttle at MRT to maintain energy for follow-on BFM. Approaching the horizontal overshoot, if the fighter assesses they will have a shot opportunity post-corkscrew, they should early-turn by executing corkscrew mechanics just prior to the horizontal overshoot and max perform in order to get the adversary into valid shot parameters. Realize that if the fighter executes this early-turn, some of the offensive advantage (nose-to-tail distance) will be sacrificed in order to kill the adversary. If the fighter assesses they would become neutral or defensive by attempting the shot, they should not take it, and continue to work to increase the offensive advantage until able to shoot and also maintain the offensive advantage. If the fighter is able to maintain the offensive position, they should continue to early-turn and shoot the adversary until a kill is achieved or until the fight is redefined.

- Neutral or defensive: If the fighter predicts the adversary will be level or nose-high at the horizontal overshoot, then the fighter is neutral or defensive. Increase turning room by staying nose-high until the horizontal overshoot (slowing downrange travel while gaining altitude). After forcing a horizontal overshoot, perform a corkscrew by reducing the throttle to idle, unloading, re-centering the stick to the most neutral position, leading with lateral stick and following through with bottom rudder, rolling to place the lift vector on the adversary. A counter-correction may be required to stop rolling and set the lift vector. If an excessive nose high attitude was required to force the horizontal overshoot, input bottom rudder to position the nose slightly above the horizon prior to executing the corkscrew. Slowly advance the throttle back to MRT and reprogram a lift limit pull. Place the lift vector on to slightly behind the adversary, when following across the bottom, maintaining a lift limit pull. The fighter's airspeed should be enough to force a vertical overshoot but not so much as to flush downrange (approximately 200-230 KIAS) at the bottom. As the adversary transitions above the horizon, assess lateral and vertical separation. If there is more vertical separation, attempt to get the nose up. If there is more lateral separation, attempt to collapse the fight by turning more level across the top. If offensive with sufficient nose-to-tail separation, then early turn via corkscrew mechanics prior to the horizontal overshoot. Put the LV in front of the adversary pulling lead to align fuselages and possibly gain a shot opportunity. However, the fighter must avoid pulling too much lead which would sacrifice positional advantage by increasing down range travel. A key determinant in winning the roller for the fighter is to strive to get the nose up when at the bottom before the adversary can get their nose down at the top, and vice versa. If the fighter can continually do this without sacrificing position, then they are then gaining the advantage needed to win in a rolling scissors. At a certain point, one aircraft will gain enough advantage to pull for a shot, or the roller will be redefined.

Fighters can also evaluate whether they are offensive, neutral, or defensive by assessing their nose position relative to the horizon when the adversary executes a corkscrew. For example, if the fighter is nose-high as the adversary corkscrews, the fighter is probably offensive.

In general, a four-step process can be used during the roller:

1. Starting from the bottom of the roller, execute a compromise pull straight up until the adversary crosses the horizon (vertical overshoot).

2. Roll to place the lift vector slightly above the horizon (about 60º AOB, 20-30º nose-high), executing a lift-limit pull until passing overtop the adversary (horizontal overshoot), or as prescribed based on offensive or defensive sight pictures.

3. Execute the corkscrew by selecting idle, unloading the jet, re-centering the stick to the most neutral position, and leading with lateral stick while following through with bottom rudder to roll and place the lift vector as required. A counter-correction may be required to stop rolling and set the lift vector. Typically, lift vector placement should be on the adversary.

4. Use a lift-limit pull while maintaining the lift vector in the desired location. Reselect MRT as the nose approaches the horizon. If trending to be slower than 230 KIAS at the bottom, then re-select MRT early.

<u>Redefining the Roller:</u>

When redefining the roller, the important thing to recall is the T-45C's smallest Split-S altitude. On average 6,000 ft above the hard deck is a good number to remember when at 150 KIAS. If the fighter lacks the altitude required to optimize our LV placement during the roller, it may be time to transition to 1C flow or a flat scissors, execute an energy rate deck transition (ERDT) to set 2C flow, or put the lift vector in lead to continue the roller.

To perform a full iteration of the roller, the T-45C needs 4,000 ft of turning room above the deck. If less than 4,000 ft above the hard deck, fighters have three options:

- If offensive (> 30º advantage): Avoid committing too nose-low, post-corkscrew by placing the lift vector closer to the horizon. This option will allow the fighter to perform a roller iteration whereas the adversary may not be able. To do this the fighter will most likely be placing the lift vector in front of the adversary. In doing so, the fighter is pulling lead and giving up turning room. Fighter's should consider this option when offensive with sufficient nose-to-tail separation. If the adversary reverses at the top, they will be setting 1C flow, so modulate the throttle and nose position to ensure airspeed is maintained within the T-45C radius band. If the adversary overbanks and executes an ERDT, be sure to arrive on deck within the T-45C rate band.

- If neutral, slightly offensive, or slightly defensive (< 30º advantage or disadvantage): Redefine into the flat scissors or set 1C flow. Keep max performing the aircraft and work for a horizontal overshoot. Once the overshoot happens, perform a nose-high reversal, striving to arrive in the radius band or flat scissors band.

- If defensive (> 30º adversary advantage): Utilize an energy rate-deck transition. Once the adversary has committed nose-high, overbank in the current direction, trade altitude for airspeed, and transition to 2C flow.

Radial G affects the "sight picture" of being offensive, neutral, or defensive at various points in the maneuver. Fighters can experience several optical illusions. At the top, slower airspeed and Radial G give the fighter a smaller turn radius (reference Figure 4-4), while greater airspeed at the bottom causes a larger turn radius. As a result, relative position of the aircraft alone does not determine actual advantage. For example, if the fighter is neutral in the fight, at the bottom of the roller the fighter is slightly ahead of the adversary, while at the top the fighter is slightly behind the adversary.

<u>2C Redefinition Recognition and Counter</u>:

For many reasons, it is critical to recognize the adversary's 2C redefinition as soon as possible, because the goal is to kill the adversary, not become neutralized. Early recognition allows the fighter to keep more airspeed and reorient the LV to establish misaligned turn circles. Generally the 2C redefinition will be initiated from the top of the roller. The fighter must lower the nose to trade altitude for airspeed while typically maintaining a pull that does not let the defender drift aft on the canopy. If the adversary's 2C redefinition is not aggressive or the defender arcs, sufficient turning room may be available to pull for a shot approaching the deck. If the defender executes sound 2C redefinition mechanics, attempt to misalign turn circles, execute a timely energy excursion, and employ weapons as described in the next section.

Learning Objectives (emphasis) for the roller:

1. PADS set up

2. 100% adherence to training rules

3. Lift vector placement

4. AOA and airspeed control

5. Flight path projection

6. Fight redefinition

505. OFFENSIVE PERCH BFM OVERVIEW

Perch BFM is partial task training to practice BFM with respect to an aircraft's relative location to the bubble. As described in earlier chapters, the average turn radius of the T-45C is 3,000 ft with a bubble of 6,000 ft in diameter. Thus, the 6k perch set starts with the attacker just outside the defender's bubble and the 9k perch set starts with the attacker well outside the defender's bubble. In the fleet, a 3k set will be practiced that starts inside the defender's bubble.

NOTE

> A 2C fight is not necessarily a pure rate fight. The less neutral the fight, the more it is a war of energy excursions, misaligned turn circles, and angles. As the fight becomes more neutral, it transitions to a true rate war.

All offensive perch sets will be executed as below with the exceptions and considerations noted for each set.

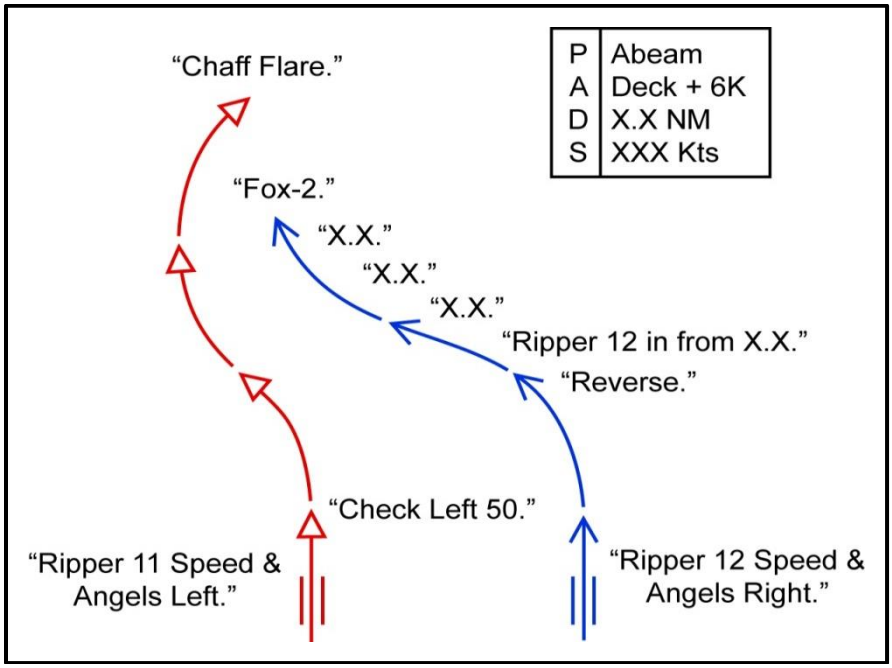


Figure 5-17 Generic Perch Setup

After a "speed and angels" call from both aircraft, Lead will check the flight 50° away from the fighter. The fighter will select MRT and turn toward the adversary using G to maintain airspeed. The fighter will direct a "reverse" when the adversary is at the fighter's canopy bow. The fighter will then reverse the turn and fly a pure pursuit curve, maintaining altitude, and putting the adversary between the airspeed and altitude boxes in the HUD. The throttle will remain at MRT for compressor stall mitigation in the T-45C. The fighter will call "Ripper 12 in from X.X" and count down the ranges in tenths of miles while directing the adversary to "tighten/ease your turn" to maintain 40° AOT. At the desired range, the fighter will call "Fox-2." If the IP deems the A/A TACAN is lagging too much, they can initiate with a "fight's on."

At the "Fox-2" or "fight's on" call, lag the defender, and drive to the point of departure. A good technique is to use a geo-reference point to drive to defender's departure point (e.g., a cloud, ground ref, etc.). When aspect stops changing and the Line of Sight (LOS) rate explodes, the attacker is inside the defender's bubble and has arrived at the AW. To execute AWE mechanics, roll to place the lift vector on to slightly below the defender and pull to stabilize the defender on the canopy (about one fist above the canopy bow). Analyze the AWE and subsequent defender maneuvering.

1. If the AWE is early, it will be easy to pull the defender forward on the canopy, possibly to the HUD. Use a lag maneuver to prevent an in-close overshoot. Ease the pull, allowing the defender to drift aft on the canopy and then reset the pull. If very early, roll up (<45°) and out of plane momentarily, then re-establish the pull with the lift vector back on the defender. An early correction normally results in a flight path overshoot and sight picture similar to a late entry.

2. If the AWE is late, the attacker will be unable to stabilize the defender on the canopy while max performing. In this case, capture airspeed in the rate band. Be patient and wait for misaligned turn circles to develop. If the attacker's AWE is excessively late, the fight will be more neutral and closer to a true 2C rate war.

3. With "nailed" AWE timing and properly executed AWE mechanics, the attacker will be able to stabilize the defender on the canopy right above the canopy bow with only a small amount of movement aft (i.e., a slight flight path overshoot). Capture in the upper end of the rate band (300-330 KIAS).

The defender will move aft on the attacker's canopy slightly during the flight path overshoot due to misaligned turn circles. Be patient, and do not bleed below 300-330 KIAS. After the flight path overshoot the defender will stabilize on the canopy momentarily before starting to move forward. This is the cue to perform an energy excursion. Place the lift vector on the defender and pull to the nibble of buffet, trading airspeed for nose position. The defender should recognize the attacker's nose coming on (TA decreasing) and perform a defensive energy excursion. If unable to counter the attacker's nose coming on due to a low energy state, the defender will perform one of two nose-low maneuvers to make use of potential energy in the form of altitude: a redefinition or a ditch.

<u>Redefinition Follow</u>:

The defender will perform an oblique nose-low maneuver if assessing the attacker's nose as a threat. By maneuvering nose low with a max performance pull, the defender is looking to regain lost energy while momentarily forcing the attacker outside of the bubble allowing the defender to momentarily take back angles. This maneuver will force the attacker to decide between weapons employment and follow-on 2C BFM. Prudent fighters should always choose follow-on BFM (OBFM Objective number one) to avoid a loss of angles upon arrival on the deck. This decision, however, will allow the defender to create angles and hold the attacker's nose in lag. This can be compensated for by driving towards the defender's point of departure where the defender first redefined. The cues for redefinition follow timing are the same in the vertical realm as they are in the horizontal. When the attacker's sees the LOS rate "explode," it is time to follow. A good technique is to look for LOS rates in reference to the ground vice the canopy since the motion

will be in the oblique. When LOS rate explodes, roll and place the lift vector on the defender and max perform to stabilize the defender on the canopy. Do not attempt to gain vertical turning room while driving to the point of departure when countering this defense as this will keep the attacker outside the defender's bubble longer. The goal is to stop losing angles to the defender by quickly entering the bubble. This concept differs from a ditch follow since those redefinitions are 1C in nature vice 2C.

1. If the follow timing is nailed (i.e., the attacker sees stable TA and canopy position from the defender), the fighter has maintained the advantage. The defender will stabilize approximately 15° below the horizon. The fighter should keep the lift vector on to slightly above the defender and be patient. The deck will quickly become a factor and MATC will provide weapons employment opportunities.

2. If the follow timing is early, the attacker will be able to pull the defender forward on the canopy – even just a small amount – and the attacker will see the defender nearer to the horizon. This will result in more angles to solve for on the deck with less of an altitude advantage to help solve for them. If left unresolved, the attacker will have an ICFPOS and arrive on the deck with many angles to work through outside of the defender's bubble. This gives the defender a reversal opportunity at altitude but also the opportunity to take back angles while the attacker is outside the bubble on the deck. To correct, the attacker must ensure that any overshoot occurs inside to slightly aft of the defender's CZ. In order to avoid an in-close overshoot, either ease the pull to stabilize the defender on the canopy or perform an aggressive lag maneuver (depending on how premature the follow). Once executed, place the lift vector on the defender and continue the lift-limit pull at MRT all the way to the deck. If the early follow is recognized and corrected for quickly, the fighter can maintain the advantage on the deck. However, time to kill will increase.

3. If the follow timing is late, attacker will see the defender move aft on the canopy when established in a lift limit pull. Essentially, the defender is out-rating the attacker. Additionally, the defender will appear farther below the horizon than on a well-executed entry. Unfortunately, those angles have already been lost. To compensate, simply keep the lift vector on the defender, and follow the defender down to the deck. Maintain the lift-limit pull at MRT, heading to the deck, capture in the upper end of the rate band and wait for MATC to develop. Assuming the following is not excessively late, MATC should develop, and an energy excursion executed to take a shot. The late follow effectively increases time to kill.

<u>Ditch Follow</u>:

A ditch can be thought of as a pure nose-low vertical 1C redefinition. The timing and cues when following a ditch are the same for the oblique nose-low out-of-plan redefinition follow with a couple minor exceptions. First, an altitude advantage will help attacker compensate for the steeper POM. Attempt to climb while driving to the defender's point of departure in order to maximize the altitude advantage. Second, attempt to match the defender's POM down to the deck.

Fighting 2C On the Deck:

Approaching the deck, the attacker should use the 10% rule to avoid busting it. A proficient fighter should initially target approximately 45-50° nose low. However, 30° nose-low could be effective. If there is any excess altitude between the defender and the deck, the defender has essentially allowed turning room, and attacker should immediately take advantage of it. If the attacker sees stable canopy position during the transition to the deck, the attacker should maintain that current sight picture but not bleed below 300 to 330 KIAS to avoid getting stuck in lag. As the attacker captures in the upper end of the rate band, MATC will develop. MATC cues on the deck are the same as at altitude following the initial AWE. If defender begins to track aft on the canopy, capture airspeed. If the defender tracks forward on the canopy, perform an energy excursion for a shot.

After each shot, the attacker should ***immediately*** maneuver to lag to preserve the offensive advantage in the defender's CZ. If not near the CZ and outside the defender's bubble, perform AWE mechanics assessing canopy motion, adjusting the pull appropriately to stabilize the defender on the canopy. After a series of these shots and maneuvers, the attacker should arrive in the near side of the defender's CZ with RAC under control. The defender will be out of airspeed, altitude, and options. The attacker needs to be cognizant of any closure problems the defender may present in attempting to force an overshoot. Modulate the throttle to maintain the advantage. If the defender reverses, transitions to a 1C fight or flat scissors. Use the vertical appropriately and control closure as required.

The overall objective is to arrive in the defender's CZ with range, angles, and closure under control. Take shots of opportunity and apply follow-on BFM (e.g., a lag maneuver back to the CZ) to maintain the offensive advantage.

506. 6,000 FT OFFENSIVE PERCH SET (6K)

Climb to PADS altitude at 300 KIAS abeam lead. To achieve 350 KIAS, lead will direct a level acceleration for the extra 50 knots. Lead will call to "set 350." Select MRT. Keeping lead on the horizon, adjust the throttle to maintain abeam and arrive at PADS altitude and airspeed. Once established, Lead will initiate the "speed and angels" call.

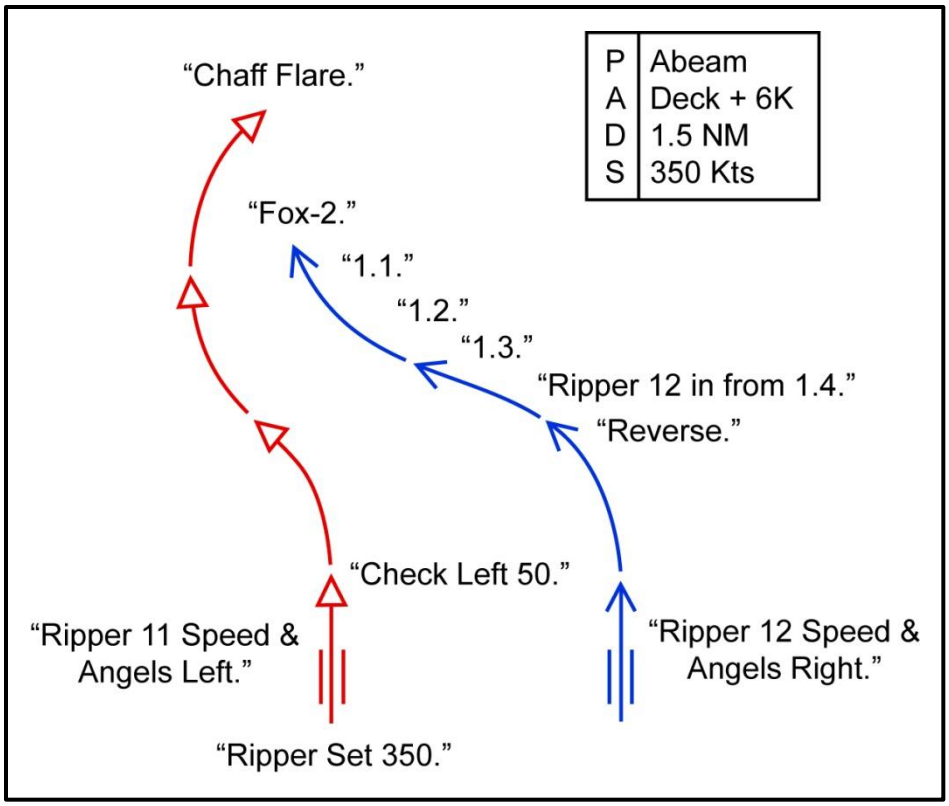


Figure 5-18 6,000 ft Perch Setup

The 6k set positions the attacker just outside the defender's bubble. Thus, after calling Fox-2 some angles will be gained by the defender while the attacker drives to the AW. Weapon selection should be primarily the AIM-9. During this set, if the attacker nails the AWE timing and mechanics, the attacker should have a small flight path overshoot in the CZ. Following the overshoot, misaligned turn circles will work in the attacker's favor yielding an opportunity for the attacker to pull for a shot and follow the defender through the deck transition. If a shot opportunity is not immediately available, capture to maintain airspeed and look for misaligned turn circles to provide an employment opportunity.

507. 9,000 FT OFFENSIVE PERCH SET (9K)

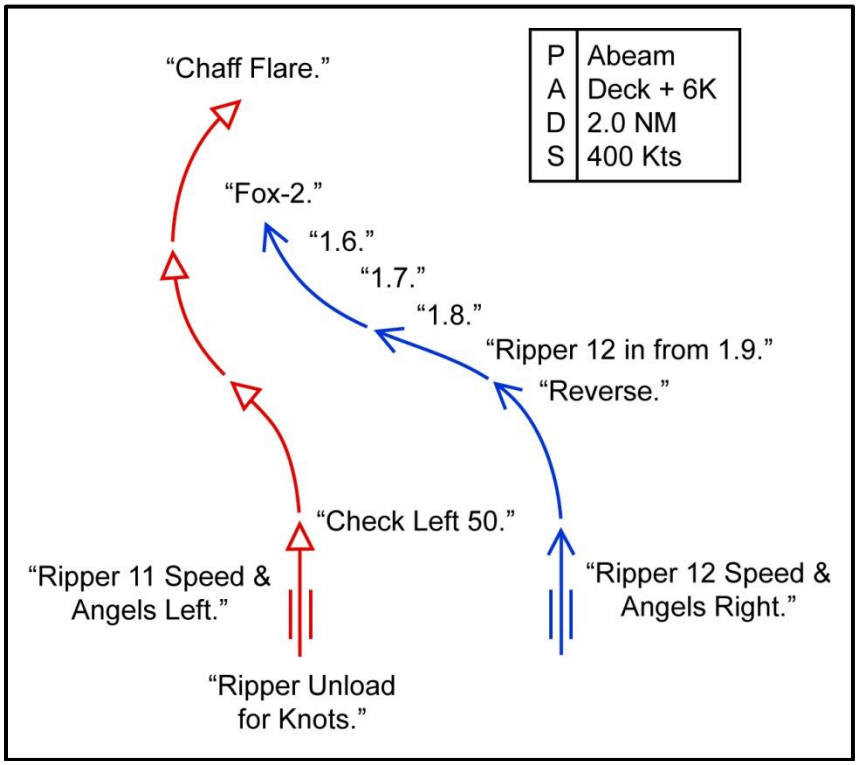


Figure 5-19 9,000 ft Perch Setup

Climb to PADS altitude + 2,500 ft at 300 KIAS abeam Lead. To achieve 400 KIAS lead will direct an unload for the extra 100 knots. Lead will call to "unload for knots". Select MRT and unload to set 7 degrees nose low. Keeping Lead on the horizon, adjust the throttle to maintain abeam and arrive at PADS altitude and airspeed. Once established, Lead will initiate the "speed and angels" call.

The 9k set positions the attacker outside the defender's bubble. While outside the bubble, the defender will be able to generate angles via target aspect change prior to LOS rate explosion. Also, due to the range and misaligned turn circles, the attacker will have two bubble entries. Weapons selection should be primarily the AIM-9. At 400 KIAS, the fighters are close to corner airspeed, so care must be taken not to overstress the aircraft. Perform a smooth pull to target 7.0 G, or a lift-limit pull, during the AWE. With a nailed AWE and the defender stable above the canopy bow, expect a flight path overshoot aft of the CZ with medium AON. Following the overshoot, deep MATC will develop, and the attacker should execute an energy excursion to re-pierce the defenders bubble (second bubble entry). If available, take a follow-on shot and immediately go to lag. If a follow-on shot is unavailable, immediately lag in order to preserve the offensive advantage and drive to the new AW. From this point, the fight should look like the 6k set with the exception of the defender having generated slightly more angles. The attacker needs to remain in the rate band and utilize MATC combined with timely energy excursions to take valid shots.

508. OFFENSIVE BFM CONCLUSION

Keys to success in OBFM (and all of BFM) at this level are to execute solid Admin and Tac Admin, adhere to training rules, and execute safely. A solid understand of basic BFM concepts definitions, and T-45C mechanics is critical to success, but the successful application of those concepts will take time and experience. Remember, the first priority is to ***maintain the offensive position*** and then kill the adversary.

Conclusion:

The key to mastering offensive BFM is understanding and solving the three BFM problems: RAC. The fighter must perform timely and well-executed AWE mechanics, utilize proper redefinition follow-on mechanics, and intelligently perform an energy excursion to kill the adversary in a timely manner. This takes solid knowledge of the concepts, sound execution, and aggressiveness.

Learning Objectives offensive perch sets include:

1. PADS set up
2. 100% adherence to training rules
3. AWE recognition and timing
4. AWE mechanics
5. Energy management
6. Fight redefinition recognition and follow
7. LAR and shot opportunity recognition
8. 100% valid shots

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CHAPTER SIX
DEFENSIVE BFM

600. DEFENSIVE CONCEPTS AND DEFINITIONS

The concepts learned in OBFM also apply to Defensive BFM (DBFM). The difference is that the defender needs to assess the BFM cues to create RAC problems for the attacker all while looking over the shoulder. The better the defender can do this, the more successful they will be at achieving the DBFM objectives.

Defensive Objectives:

1. Defeat initial weapons employment.

2. Maneuver to deny follow-on weapons employment.

3. Neutralize the fight.

4. Transition to HABFM.

NOTE

HABFM will not be executed until the first HABFM event.

Defensive maneuvering is hard work. It is physically demanding and mentally challenging to assess the visual BFM cues presented by the attacker. While reading the following, remember that the defender cannot fight what they cannot see. In order to achieve the defensive objectives.

The defender must maintain sight.

There is a saying in the strike-fighter community, "lose sight, lose the fight." This concept cannot be overemphasized. If the defender loses sight of the attacker, they will be unable to defend against shots.

Sensor Nose:

In the fleet, the definition of "sensor nose" will change depending on the threat aircraft's radar and other advanced systems. It is likely that sensor nose may be greater than 45° off the actual nose position of one's opponent. For our purposes while training in and against the T-45C, we will define sensor nose as "nose on."

To execute your defensive BFM objectives, the defender must do the following:

1. Keep sight or regain sight – The defender cannot fight what they cannot see. However, the defender will lose sight at times just as a function of geometry (attacker passing the defender's extended 6 o'clock when establishing MATC). If sight is lost, the defender must visualize the

geometry of the fight and bias their scan to where the attacker should reappear. The defender should be ready to execute the "Lost Sight Game-plan" found later in this section.

2. Max perform the aircraft – Don't just float the turn and become a target.

3. Avoid the deck – This was a challenge during your Offensive BFM training, but it's now more difficult due to the increased time spent looking over the shoulder. A rapid inside-outside scan is a major key to success. Being able to fly close to the deck gives the adversary less turning room to solve the RAC problem.

Above all, no matter how dire the situation may seem, never give up!

Defensive Axioms:

1. Survive:

   a. Deny sensor nose.

   b. Defeat shots.

2. Attacker moving forward on canopy = Keep pulling.

3. TA decreasing = Increase pull to hold attacker's nose off.

4. If the attacker is <60º AOT and unable to perform steps 2 and 3, redefine.

Lost Sight Game-plan:

If lost sight on the deck, max perform in the current turn direction for 3-5 seconds. If the attacker's nose is in lag and aft of the defender's visual limit, a max performance pull should make a valid weapons solution more difficult and create some fuselage misalignment. After 3-5 seconds, rapidly reorient the lift vector to pull in the opposite direction and max perform. Sight should be regained at the high right 5 or left 7 o'clock scanning high to level (possibly low if the defender climbed). If the attacker is positioned just aft of the visual limit, a max performance pull should pull the attacker forward and within sight of the defender. If the attacker's nose is in lead pursuit and is unseen by the defender, a reversal will effectively solve the attacker's RAC problem and result in a shot. ***In all cases, be prepared to defend!*** Execute a missile defense or guns-d as appropriate.

If lost sight with excess altitude above the deck, continue in the current turn direction and max perform nose-low to the deck. If the attacker is outside the defender's bubble, the defender should gain angles. If the attacker is on the cusp of the defender's bubble, the max performance pull should momentarily push the attacker outside the defender's bubble and gain angles while taking back angels and giving the attacker a RAC problem to solve. If the attacker is inside the bubble approaching the CZ, the nose low maneuver will complicate the attacker's RAC problem while regaining energy for follow on DBFM. In all cases, scan for the attacker on the inside of the turn from the aft visual limit to the wing line and be prepared to defend.

6-2 DEFENSIVE BFM

Reversals:

Reversal opportunities come as a result of overshoots. The question of whether to reverse or redefine the fight can be difficult to answer. A defender must recognize the circumstances leading to the overshoot and the type of the overshoot witnessed in order to know if a reversal is warranted.

- 3/9-Line Overshoot – If defensive and the attacker has moved forward of the 3/9-line, a role reversal has occurred. Transition to OBFM.

- Flight Path Overshoot – If the defender observes a flight path overshoot where the attacker is in the CZ or aft, do not reverse. A reversal will merely help the attacker's lag problem and aid in the defender's demise.

- In-Close Flight Path Overshoot (ICFPOS) – How close is "in-close?" If the overshoot occurs inside the forward limit of the CZ, it is considered it "in close." Also needed is high Target Aspect (> 60° TA). If the defender does an immediate, aggressive reversal after the overshoot, it may induce a 3/9-line overshoot from this flight path overshoot.

In order for the fighter to take advantage of the adversary's overshoot, three criteria must be satisfied:

1. The adversary must have an ICFPOS (< 2,000 ft) with greater than 60° TA.

2. The adversary must pass the extended six o'clock with a high line-of-sight rate.

3. The defender must be able to visualize reversing inside the adversary's turn radius. Know these criteria like boldface.

If ever in doubt, do not reverse.

Reversal Technique:

With ICFPOS criteria met, unload, roll, place the lift vector on to slightly above the adversary, and perform a lift-limit pull, trading airspeed for angles and altitude to transition to scissor maneuvering. The fighter is now pulling into a flat scissors, so airspeed must be bled down from 300 KIAS to reach the optimal flat scissor performance numbers. A break turn (21 units AOA), will bleed airspeed quickly while quickly establishing a nose high attitude. If already slow, a nibble of buffet pull might help the nose smoothly track up and preserve some energy for the follow-on BFM. Lift vector placement is crucial. The lift vector needs to be on the adversary (to slightly high) in order to collapse the fight and flush the adversary forward of the 3/9-line. Update the LV to give the adversary as little turning room as possible. The use of speed-brakes is a technique that is often introduced here. The fight has now transitioned from a rate fight to a positional fight. Apply techniques discussed later in the flats and roller sections to gain an offensive advantage.

601. SNAP SHOT DRILL (SSD) AS THE TARGET

The defender's goal during the SSD is to recognize what an impending gunshot looks like during the non-maneuvering sets and then be able to defeat the shot by executing a proper guns d during the maneuvering sets.

Execution:

The target aircraft, you will tighten or ease G initially to maintain the shooter at the 10 or 2 o'clock position and then allow the shooter to drift aft. During the maneuvering sets, the target should evaluate the nose position of the shooter and execute a properly timed guns d to defeat the impending gunshot. The typical flow for a defensive SSD will be two passes as a non-maneuvering target followed by four passes as a maneuvering target.

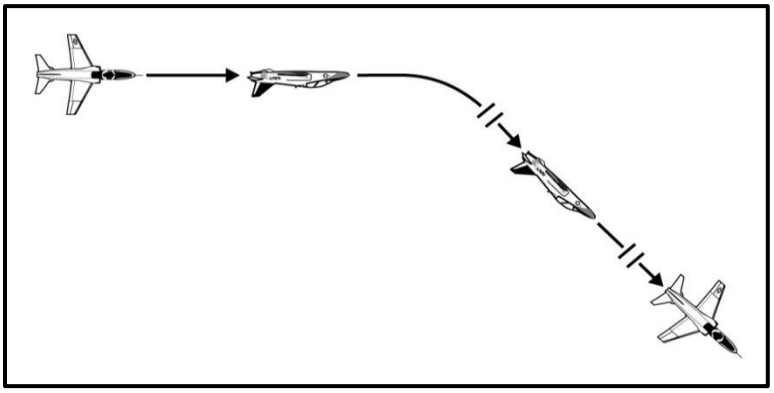

Figure 6-1 Guns Defense

Performing the Guns Defense (Defeating Plane of Motion):

The target during the maneuvering sets, has two goals. First, the target must present the shooter with as little surface area as possible by placing their wingtip on the shooter. Second, the target needs to maneuver out of plane in a timely manner to deny the shooter the ability to solve for plane of motion.

Timing:

It is imperative that the defender time the guns d to defeat the shot. If the maneuver is too late, the target will fly through the shooter's pipper. If the maneuver is too early, the defender is broadcasting their intentions, allowing the shooter to counter the maneuver and solve for POM. The target aircraft must develop a sight picture of the shooter's nose position during the "trigger down snap" call on the non-maneuvering sets and then use this evaluation to time the guns d.

Maneuvering:

Three basic maneuvers can be performed to defeat impending gun shots:

- = Goods
- o = Others

1. Near wingtip on, max performance pull nose high –
   - May be the only option when on the deck.
   - Quick to perform and easy for target to maintain sight. Allows for good aircraft displacement and good follow-on BFM.
   - o Maneuvering against gravity.
   - o Need some airspeed (energy) to displace aircraft.

2. Near wingtip on, bunt –
   - Quick and may fake the shooter into going nose-high.
   - Easy to maintain sight.
   - o Little displacement of aircraft from the current POM and may accelerate farther out in front i.e., poor follow-on BFM.
   - o Not an option when fighting less than 500 ft above the deck.
   - o The amount of negative G needed to displace the aircraft can be extremely uncomfortable.

3. Far wingtip on, max performance pull nose-low –
   - Best aircraft displacement due to increased G available from gravity.
   - o May broadcast intentions.
   - o Hard to maintain sight and may cause disorientation.
   - o Minimal options for follow-on BFM.
   - o Not an option when fighting on the deck.

When attempting to max perform, it is imperative that the target pull to the lift limit. If the pull is too hard (past lift limit) or too weak, the target will not achieve maximum aircraft displacement. For the T-45C, lift limit can be achieved by a rumble of buffet pull.

Safety:

If the target assesses the shot will be inside 1000 ft, a “skip it” call shall be made, and the target will ease their pull. If it appears the 500 ft bubble will be broken, the target will de-conflict by maneuvering up and away, and call “knock-it-off.”

602. DEFENSIVE FLAT SCISSORS

The flats entry and mechanics will remain similar to offensive BFM. The adversary will, however, begin moving to an offensive position early (on the “2” of “3, 2, 1”) so that at the “fight’s on” call, roles are clearly established. The defender will be looking to deny shots, defeat shots, or redefine, in order to denying access to the CZ. Survival in this defensive fight depends on the defender’s ability to recognize whether the fight is a true flat scissors are closer to a 1C fight and apply the appropriate timing and mechanics.

Reversal Timing:

If the adversary places their lift vector on the defender, the defender must honor the adversary's lift vector by placing their lift vector on the adversary. This will rapidly increase closure and decrease range thereby giving the adversary the maximum RAC problems to solve. If the adversary executes a timely early turn, the defender will need honor the early turn with their own reversal to complicate the adversary's RAC problem (turning lag in to lead) and deny access to the CZ. In general, time defensive reversals with the following criteria in mind:

- To work out of phase, the defender should reverse after the adversary crosses their extended six. This will help increase the lateral separation needed to work towards a high-aspect or neutral pass (1C fight).

- To work in phase, the defender should reverse when the adversary reverses. This will help decrease lateral separation and align fuselages in working towards a position underneath the adversary (flat scissors).

- If the adversary reverses in front of the defender's 3/9-line, transition to OBFM.

Defensive scenarios and strategy can be defined with respect to energy and position:

- Up and forward – The defender should work to get out of phase and attempt to stack the flats to deny the adversary a WEZ and work towards a neutral slow speed merge. If the adversary is working into a weapons envelope, look to redefine or execute a guns d.

- Down and forward – With a small amount of nose-to-tail separation, the defender should attempt to work in phase below the adversary, potentially masking the adversary's view of the defender. If the adversary has a good amount of nose-to-tail separation, the defender should redefine into a 2C fight.

Redefining Nose-Low:

If the adversary is able to gain weapons separation for a snap shot or missile employment, or has worked in phase aft approaching the defender's CZ for a possible tracking solution, it is time for the defender to get out of the flat scissors and redefine 2C. In CNATRA training, two valid snap shots equal a kill as does one second of tracking time. After a guns d or missile d, execute a nose low redefinition. If as a result of the shot attempt the adversary has an ICFPOS, the defender should defend and stay in the flats in order to neutralized the attacker or possibly transition to OBFM.

Redefinition scenarios and strategy can be defined with respect to energy and position:

- Up and forward – Honor sensor/gun nose with the appropriate defense and execute a pure ditch (assuming approximately 6,000' above the deck starting at 150 KIAS). Roll to place the lift vector on to slightly in front of the attacker and max perform

through pure nose low. This will force the attacker to drive to the defender's point of departure and execute a ditch follow. The defender must see the attacker's follow and reoriented the lift vector back into the attacker to set 2C flow.

- Down and forward – Honor sensor/gun nose with the appropriate defense and execute a nose-low out-of-plane 2C definition (as discussed in the next section). A pure ditch in this case would allow the attacker to drive straight in to the defender's control zone.

Learning Objectives (emphasis) for the flats:

1. PADS set up
2. 100% training rules adherence
3. AOA and airspeed control
4. Lift vector placement
5. Recognition of offensive and defensive position
6. Reversal timing execution
7. Shot opportunity recognition and mechanics
8. Guns d recognition and mechanics
9. Fight redefinition

603. DEFENSIVE ROLLING SCISSORS

Redefining a fight into a roller can be advantageous for a defender. If executed properly, the adversary will have to solve several BFM problems prior to being able to take a shot.

Execution:

The execution and maneuvering of the defensive roller will remain the same. The adversary will, however, maneuver to establish an offensive position prior to "fight's on." Attempt to gain a positional advantage by adhering to the 4-step process. Reference neutral or defensive mechanics from the offensive roller discussion. Ultimately, the defender's goal is to take back ground in the roller or, if necessary, redefine the fight prior to the attacker's sensor or gun nose becoming a factor.

Redefining the Roller:

- Slightly defensive (< 30º adversary advantage) – Without sufficient altitude and turning room to perform another roller iteration, the defender should redefine the

fight into either 1C flow or a flat scissors. To achieve this, the defender will max perform approaching the top of the roller to work for a horizontal overshoot. Once the overshoot occurs or the attacker's nose is sufficiently in lag generating as much misalignment as possible the defender should perform a nose-high reversal.

- Defensive (> 30º adversary advantage) – If definitively defensive, the defender is effectively in a 2C fight. If the defender assesses the attacker will have a weapons employment opportunity after the next corkscrew, the defender should execute an ERDT in order to hold the attacker's nose in lag while simultaneously accelerating towards the rate band. To achieve this, the defender will wait for the attacker to commit nose-high from the bottom of the roller, overbank, and execute their best 2C defensive game plan as discussed later in perch BFM.

Learning Objectives (emphasis) for the roller:

1. PADS set up
2. 100% adherence to training rules
3. Lift vector placement
4. AOA and airspeed control
5. Flight path projection
6. Fight redefinition

604. DEFENSIVE PERCH BFM OVERVIEW

Defensive perch BFM sets are the same as the offensive perch sets, but with the roles reversed.

Execution:

After a "speed and angels" call from both aircraft, lead will check the flight 50° into the defender. The defender will select MRT and use G to maintain airspeed to pull to the new heading and reverse, when directed. The defender will reverse the turn and maintain airspeed and altitude, pulling the adversary to 40° AOT. The throttle will remain at MRT for compressor stall mitigation in the T-45C. The adversary will call "[tactical call-sign] in from X.X" and count down the range in tenths of miles while directing the defender to "tighten/ease" the turn to maintain 40° AOT. At the prescribed range, the adversary will call "Fox-2."

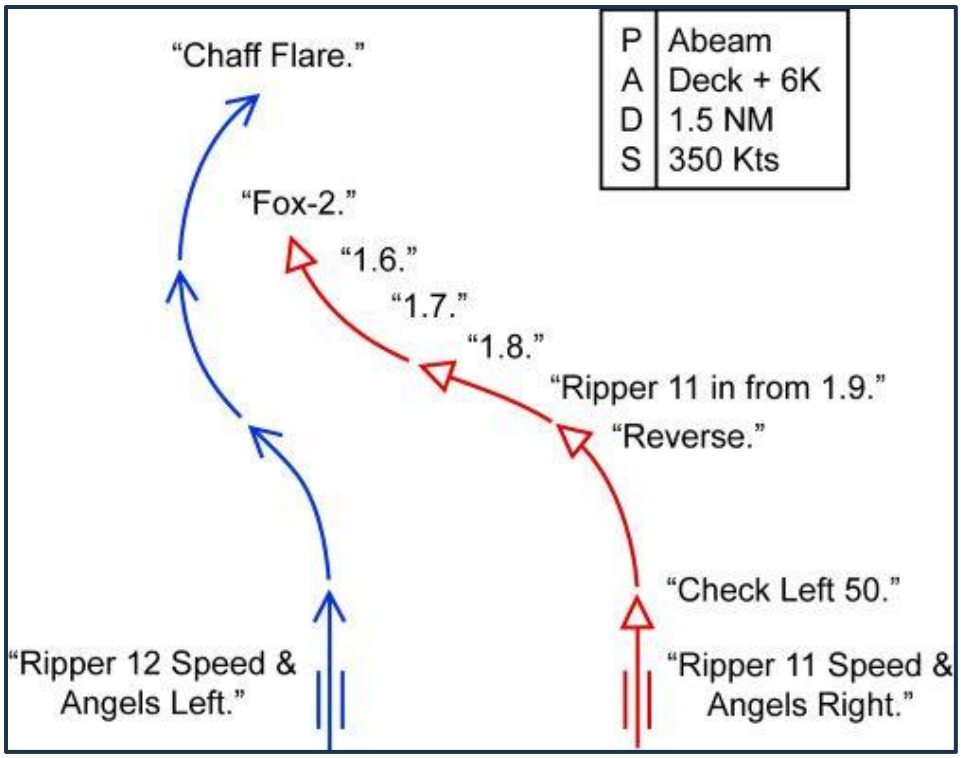

Figure 6-2 Defensive Perch Setup

At the "Fox-2" call, the defender will roll to place the lift vector on to slightly below the attacker, and perform a lift or G limit pull while simultaneously calling "chaff flare" (Axiom 1: survive). A proper defensive break turn should keep the attacker the same distance above the horizon throughout the defensive break turn. Next, the defender will assess the attacker's AWE:

1. Early AWE (decreasing target aspect) – The attacker will appear to be nose-on and the defender will be able to hold the attacker stationary or pull the attacker forward on your canopy (Axiom 2: attacker moving forward on canopy = keep pulling). If able to pull the attacker forward of 60° AOT, regardless of sensor nose, the defender should keep pulling. Call "chaff, flares" denying a missile shot, and assess if a guns d is required. The defender is defeating very low P_k shots from the attacker, while attempting to force an ICFPOS, effectively neutralizing the fight. If the attacker does not maneuver to fix the early AWE, this is good. The defender should continue to max perform down to gun range, creating an angle and closure problem, for the attacker. After honoring gun nose with an appropriate guns d, the defender should reorient the lift vector back into the attacker, looking to force an in-close overshoot. If the attacker does not maneuver sufficiently to lag, the defender will force an in-close overshoot, and the defender should now apply your reversal mechanics. If the attacker prevents the overshoot by performing a nose-high maneuver, the defender has two options:

   a. With vertical airspeed, the defender should match the attacker nose-high to collapse the range.

   b. Without vertical airspeed, continue in the current turn, 5-10° nose-low, and execute follow-on DBFM looking for opportunities to regain energy.

2. Late AWE (increasing target aspect) – The defender will be able to create greater fuselage misalignment and thus increase time until the attacker's sensor nose will become a factor. The defender should continue the max performance pull to capture airspeed in the tactical rate band. Assess AOT and TA:

a. With high AOT and high TA, the fight is a 2C rate war.

b. If low AOT and high TA, the attacker will still be able to benefit from MATC. The attacker will have a flight path overshoot. The greater the TA at the time of the overshoot, the longer it will take for the attacker's nose to threaten the defender. During the overshoot, the defender will lose sight of the attacker disappearing behind the aft visual limit as the attacker is deep in MATC. Prior to losing sight, if unable to continue to pull the attacker forward on the canopy, the defender should capture to maintain airspeed. MATC is working for the attacker when reappearing at the defender's aft visual limit. If the attacker shows decreasing TA, then the defender must execute a max performance pull. (Axiom 3: TA decreasing = increase pull to the defender is able to hold the attacker's nose off) and redefine if redefinition criteria is met (Axiom 4). If the defender is able to hold the attacker's nose in lag, they should capture airspeed and watch for MATC to develop again.

c. If the attacker is so late to the AW that the fight becomes neutral/cross-circle, the defender has transitioned to HABFM. Initiate a nose-low out-of-plane maneuver cutting across the circle. Left un-countered, the fighter will gain an offensive advantage. If the adversary counters, the fighter will arrive at a neutral merge on the deck or perhaps in a rolling scissors.

3. Nailed AWE (stabilized target aspect, low AOT) – The defender will only be able to pull the attacker forward while the attacker is outside the bubble (Axiom 2). Once the attacker enters the defender's bubble, the defender will be unable to continue pulling the attacker forward on the canopy and the defender will see stabilized target aspect as the attacker executes the AWE. The defender must capture airspeed. The attacker will have a small flight path overshoot in, or slightly aft of, the CZ. When the attacker reappears, TA will be decreasing, and the defender must max perform in order to hold off sensor nose (Axiom 3: TA decreasing = increase pull to hold attacker's nose off). If it looks like the attacker's nose is on, the defender was late to recognize decreasing TA and must redefine with redefinition criteria met (Axiom 4).

<u>Redefinition</u>:

If the defender's is unable to pull the attacker forward of 60º AOT and sensor nose is becoming a factor (attacker's TA is decreasing) the defender must redefine. The most common mistake is to be late to call "chaff flares" and/or be late to redefine. Avoid this by redefining before sensor nose becomes a factor. The sight picture for this is when the attacker has about 30° of turn to go before coming nose on in the T45C. Once the redefinition has been initiated, the defender is forcing the attacker to choose between weapons employment and follow-on BFM. If lost sight, it is better to redefine early than late.

The redefinition is an oblique nose-low out-of-plane maneuvering designed to minimize the aircraft turn radius, to force the attacker momentarily outside of the bubble, and to gain energy for a 2C fight on the deck. The effect of minimizing the aircraft turn radius allows the defender, to observe forward movement of the attacker on the canopy. This is desirable since anytime the attacker is moving forward on the canopy, the defender is taking angles back. If the attacker does not execute a proper follow, the defender can either gain angles, energy, or both, significantly increasing time to kill or perhaps neutralizing the fight.

The overall goal of the redefinition is to increase time to kill by attempting to hold the attacker on the edge of the defender's bubble and by gaining energy for a rate war on the deck. This potentially prevents the attacker's nose from coming to bear, denying a weapons solution for a greater length of time. While a redefinition will not typically neutralize an attacker, it will significantly increase time to kill. Hopefully, this will allow enough time for the defender's (notional) wingman to come help, or for the attacker to commit a BFM error. If the attacker fails to follow the redefinition correctly, more angles may be gained or a shot can be denied entirely.

The advantages of the redefinition in the T-45C are that it allows the ability to add energy even while using a lift limit pull and to maintain sight. Due to T-45C performance and visibility limitations, ***the nose-low out-of-plane 2C redefinitions is the primary defense and shall be used on the perch sets.***

To execute the redefinition:

Initially, perform a 135° overbank while maintaining a lift-limit pull (loaded rolls are ok) with the throttle at MRT targeting 45-50° nose low. This will force the attacker outside of the defender's bubble, allowing the defender to gain some angles back. Additionally, by maneuvering out-of-plane, it will displace the defender below the horizon enough to force the attacker to choose between employing a weapon and follow-on BFM. This idea cannot be overstated enough. A weak overbank and pull will result in the attacker being able to employ a weapon without sacrificing follow-on BFM. If the defender is visual at the initiation of the overbank, the attacker should be two to three fists above the head-box, and forward of the opposite canopy spike. The defender should be able to see the attacker, for example, in a left turn by placing their head to the right of the ejection seat head-box, and by looking up and aft. The attacker will typically be above the defender's altitude after the initial overbank, thus above the horizon. The defender should place the lift vector on the attacker and assess their follow.

If no joy, a mechanical 135° overbank will help pull the attacker forward and above the aft visibility limit. This should allow the defender to regain sight.

After the initial overbank and setting of the lift vector, the defender must assess the attacker's redefinition follow. The main objective is to overbank enough to force the attacker into lag, but not so much where the defender is fighting the deck (deck save) and not the attacker. Maintain a maximum performance pull at MRT and attempt to hold the attacker in lag as long as possible, arriving on the deck with energy to survive, preferably within the upper end of the 2C rate band.

1. If the attacker is early, the defender will see decreasing TA and forward LOS rates. This is due to the attacker's pursuit curve and is similar in concept to an early level AWE described

previously. Because the attacker is early and the defender is able to generate forward LOS rates, the defender should be able to stabilize the attacker on the canopy while maintaining a maximum performance pull at MRT. Use chaff/flares and a guns d as required and attempt to force an ICFPOS. If the attacker fixes the early follow, by maneuvering to lag, the defender can re-focus on executing an efficient deck transition while regaining energy while the attacker is aft of the aft visual limit. From here, it will look similar to a late follow.

2. If the attacker is late, defender will see increasing TA and forward LOS rates. This comes from the fact that the defender is "out-rating" the attacker and the attacker is outside the defender's bubble. Once the attacker turns to follow, continue to max perform and maintain the lift vector on the attacker. Because the attacker was late, the defender will see the attacker further forward than a nailed entry with greater TA, and the defender will have more energy for the rate war on the deck. All of these cues mean the defender is gaining angles back from the attacker. If the defender arcs (due to a weak pull) and accelerates well above the rate band (greater than 350 KIAS), this airspeed excess can be turned into instantaneous turn rate (via a max performance pull), creating an overall increase in turn performance.

3. If the attacker nailed the follow, the defender will see stable TA. The defender should maintain lift vector and stabilize the attacker's position on the canopy while maintaining a maximum performance pull at MRT. One the attacker passes the extended 6 o'clock, the defender should re-focus on the deck transition while maintaining a max performance pull. The defender will have about five to ten seconds to manage the redefinition before the deck will become a factor. A quick inside-outside scan is critical to maintaining SA to the deck and to intercept the 10% rule. Expect to have 330 to 350 KIAS on the deck.

<u>Deck Transition and Fighting on the Deck</u>:

As the deck becomes a factor, the defender will have to decrease AOB to avoid busting the deck. The 10% rule is the best tool for deck preservation, and it is recommended to target stabilizing approximately 500 ft above the deck when defensive. This will provide options, minimize turning room above the deck and allowing the defender to spend more time looking outside at the attacker vice hawking your altitude.

The redefinition will result in 2C flow on the deck. All the concepts and sight pictures that were discussed at the initial break turn apply on the deck now. If the attacker eases and tracks aft on the defender's canopy (to set MATC), the defender should capture in order to conserve energy on the deck. If the attacker tightens down and the target aspect starts to decrease, the defender should tighten down as well (Axiom 3). If the defender succeeds at keeping the attacker on same spot of the canopy, the attacker is effectively stuck in lag and is on the defender's turn circle. Being in the rate band would be ideal, but the defender does not have the freedom drive the fight, as they must execute defenses and energy excursions in accordance with the defensive axioms to survive (based on the attacker's actions). That being said, the defender must have realistic expectations. Even with sound execution from the defender and assuming the attacker does not make an error, the attacker will eventually arrive in a position to employ a weapon. Thus, the defender must never give up and always strive to survive as long as possible!

Ditch:

The ditch is a pure nose-low maneuver that uses precise lift vector placement to complicate the attacker's BFM problem. A ditch is useful in aircraft that have a very small vertical turn radius, using it as an advantage in an attempt to neutralize the fight. A ditch in the T-45C is challenging due to the large turn radius and poor aft visibility. As such, ditching will be used only sparingly in the T-45C but will play a significantly larger role in your fleet aircraft.

The only time a ditch is a valid BFM move in the T-45C is when the attacker is well inside the defender's bubble (e.g., redefining out of a very defensive flat scissors). If altitude is available:

1. Perform a Defensive Break Turn (DBT) to collapse the range and create as many angles as possible. It is optimum to have airspeed below 270 KIAS.

2. Unload, roll (start by simply rolling inverted), and place the lift vector slightly in front of the attacker. Set a lift limit pull and assess the attacker's follow. Throttle should be set to manage the BFM problem. If a closure problem is desired, select idle and consider using speed-brakes. If airspeed is desired for follow-on BFM on the deck, select MRT. Downrange travel will be greater with the throttle set to MRT, thus idle creates a closure problem which could result in an in-close overshoot if the attacker does not recognize it.

3. As the nose comes up to the horizon, roll to place the lift vector on the attacker to collapse the fight. If lost sight, turn in the same direction as the first ditch. These reversals are used to turn the attacker's lag into lead. Keep in mind, altitude loss in the T-45C will be excessive (approximately 6,000' at 150 KIAS) with each ditch. In order to execute two consecutive ditches, the maneuver would need to be initiated at the top of a typical MOA in order to not bust a 10,000 ft AGL hard deck.

An energy rate or positional deck transition typically follows a ditch. Ditches and positional deck transitions are 1C types of defense and involve reversals to maintain 1C flow, creating a fuselage misalignment and RAC problems for the attacker. An ERDT typically results in neutral, to slightly defensive, 2C flow.

Deck Transitions Following a Ditch and Fighting on the Deck:

If the defender does not have altitude above the deck to perform a pure nose-low maneuver (i.e., ditch), they will have to perform a transition in the oblique to prevent busting the deck. First, the defender will need to assess relative fuselage alignment. When arriving on the deck, fuselage misalignment translates into greater AOT.

1. With ***low*** fuselage misalignment (low AOT and low AON – i.e., defensive), perform a Positional Deck Transition (PDT). This maneuver trades altitude for angles and uses reversals to force 1C flow. Roll to place the lift vector below the horizon to intercept the 10% rule. Utilize a lift-limit pull with the objective of creating angles and fuselage misalignment while attempting to turn the attacker's lag into lead. As the reversals, execute a guns d if required. A closure problem may be the only way to neutralize the attacker so idle/speed-brakes may not be required.

Transition to HA or OBFM if appropriate. A PDT will result in a 1C fight or flat scissors on deck. **A true PDT in the T-45C is a BFM error as the aircraft is incapable of maintaining 140-180 KIAS (radius band) in a downhill fight. Even with low fuselage misalignment, the defender should redefine downhill in a manner that will give the attacker a RAC problem to solve and then (as airspeed builds) set 2C flow with goal of forcing the attacker's nose in to lag.**

2. With ***high*** fuselage misalignment (high AOT and high AON – i.e., the defender anticipates arriving at a relatively neutral merge with little altitude delta), perform an ERDT. This maneuver trades altitude for airspeed to ensure the defender arrives on the deck within the rate band. Target the 10% rule when committing nose-low. Update the lift vector and pull to arrive on the deck in the rate band. Greater airspeed is better, but don't ease the pull and allow the attacker drift aft. Usually 30-40° nose-low is a good rule of thumb to achieve a good ERDT. Transition to HABFM once on the deck.

3. If the defender is in defensive 2C flow following a ditch, PDT, or ERDT, execute the 2C defensive game plan outlined in the section above.

4. If the defender has transitioned to HABFM, execute the appropriate HABFM game plan (discussed later).

605. 6,000 FT DEFENSIVE PERCH SET

The 6K set positions the attacker just outside the defender's bubble. During the DBT, defender will be able to generate angles as the attacker drives to the AW. The defender must continue to max perform until they are no longer able to pull the attacker forward on the canopy and then capture airspeed. If the attacker nails the AWE, there will be a small CZ overshoot. Shortly thereafter, the attacker will execute an energy excursion to take a shot. The defender must regain sight, recognize the decrease in TA and match with an increase in pull to hold off sensor nose. When redefinition criteria are met, redefine.

Learning Objectives (emphasis) Defensive Perch sets:

1. PADS set up

2. 100% adherence to training rules

3. Defensive break turn mechanics (lift vector placement)

4. AWE timing recognition

5. Sensor nose recognition

6. Fight redefinition selection and mechanics

a. 2C

b. Ditch

c. Positional

d. Energy Rate

7. Deck transition

8. Energy management

9. Reversal criteria recognition

606. DEFENSIVE BFM CONCLUSION

Defensive BFM is extremely difficult and sometimes frustrating. A solid understanding of the aircraft performance capabilities and BFM concepts will, however, give the defender tools needed to survive. This section has described many techniques in an effort to capitalize on the mistakes the adversary may make. The fighter will need to keep one thing in mind when defensive: ***never give up***. The fighter may be able to turn the tables and kill the adversary, stay alive long enough for a wingman to assist, or perhaps disengage and live to fight another day.

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CHAPTER SEVEN
HIGH-ASPECT BFM

700. HIGH-ASPECT CONCEPTS AND DEFINITIONS

HABFM will call upon aircrew to use all the skills learned thus far in perch BFM. The key to being successful in HABFM is for aircrew to recognize what position they are in, the position in which they are likely to become, and fighting their best fight. This will take time, patience, and a continuous assessment of angles and energy. The basics of LV placement, energy management, and controlling merges will aid aircrew and are key to being successful; no matter how well those basics are executed, the fight may quickly transition to offensive or defensive BFM, allowing aircrew to then apply the principles discussed earlier to kill the adversary, or to live to fight another day.

HA Objectives (in order):

1. Deny adversary weapons employment.

2. Achieve first weapons employment.

3. Gain a positional advantage, transition to OBFM.

4. Employ follow-on weapons.

5. Timely transition to DBFM, if required.

HA Basics (tools to become successful):

1. LV placement combined with out-of-plane maneuvering.

2. Airspeed excursions (energy management).

4. Controlling merges.

5. Bubble Relationships

In order to achieve these objectives aircrew need to properly utilize the basics of HABFM.

Lift Vector Placement:

Wen maneuvering out-of-plane with respect to an adversary, the fighter's pursuit curve is defined by its lift vector placement. A good starting point is to place the lift vector on the adversary (pure pursuit) and then vary to either lead or lag.

Pure = Deny the taking of angles by the adversary and potentially force a BFM error.

Lead = Capitalize on an uncountered out-of-plane maneuver, pierce the adversary's bubble, and quickly employ weapons.

Lag = Gain exclusive use turning room (i.e., maneuvering pure nose high when the adversary does not have vertical airspeed).

Energy Management:

Fighters need to manage energy properly to ensure the ability to utilize airspeed excursions when needed. In the T-45C, aircrew can manage and preserve energy utilizing the following performance pulls:

1. Max performance pull (21 units AOA or 7.0 G)

2. Compromise pull (17 units AOA or target 1 to 2 G less than maximum performance)

3. Unload (7 units AOA or 0 G)

Airspeed Excursions:

Fighters need to be judicious in employing energy excursions. Due to the poor thrust-to-weight ratio of the T-45C, aircrew have limited ability to regain lost energy. That being said, aircrew need to trade airspeed for angles or nose position ***only*** when attempting to achieve the following:

1. Take a shot.

2. Deny a shot.

3. Take or create exclusive-use turning room.

4. Deny exclusive-use turning room.

Controlling Merges:

Controlling every merge is absolutely critical to winning a HABFM engagement. The fighter must assess altitude, energy, geometry, take out the turning room, and set the flow. These factors must be considered pre-merge (as early as the brief) and then updated after the adversary makes their initial move (as the adversary always gets a vote).

1. Altitude – Fighter should look to utilize altitude to their advantage. If near the deck and the fighter is stacked low, the adversary is unable to commit nose-low, so the altitude to the deck becomes exclusive-use turning room for the fighter. If high and slow, it is best to arrive at the merge stacked high with a lateral component which will yield a positional and energy advantage to the fighter. In training, fighters must also consider their proximity to top of the MOA or the hard deck when considering a course of action.

7-2 HIGH ASPECT BFM

2. Energy – Fighters must assess their energy state in relationship to the adversary. In the T-45C, a low energy adversary will have a slightly elevated nose attitude (increased AOA) relative to their path of motion. Fighters must also consider their energy trend in relationship to the adversary (i.e., a nose-high fighter is losing energy while a nose-low fighter is gaining energy).

3. Geometry – Angles equal a BFM advantage. Fighters should look to choose merge geometry that is advantageous (allows the taking of angles) to their intended follow-on flow. In general, a fighter looking to set 2C flow should set lateral merge geometry and a fighter looking to set 1C flow should stack the merge below.

4. Taking out the Turning room – Turning room belongs to whichever fighter takes it. If pre-merge turning room exists, tighten down to take it out (IAW training rules) which will yield an angular advantage at the merge. Post merge, as LOS rates take off, max perform across the adversary's tail to create a flight path crossing in the CZ. This max performance pull will take out any turning room from the adversary and make it easier to maintain sight post-merge. Approaching a stacked merge from below with the intent to maneuver nose high, look to gain angles underneath the adversary (IAW training rules) as this will keep the adversary further forward on the canopy and aid in maintain sight.

5. Set the Flow – When deciding whether to set 1C or 2C flow, the fighter must assess all of the factors above. When not considering the unique merges (discussed later) in the T-45C, the most relevant factors are turning room and angles as energy is not easily regained in T-45C and in the level plane, remain relatively equal. In general, if able to take angles pre-merge, the fighter should continue to exploit that angular advantage post-merge with the flow selected. For example, if able to take angles pre-lateral merge, the fighter should continue in 2C flow across the adversary's tail and NOT reverse 1C to yield lateral turning room to the adversary. If angles are gained in a stacked merge underneath the adversary, the fighter should consider reactively maneuvering 1C nose high to drive the fight toward the radius band, minimize turn radius relative to the adversary, and gain a positional advantage. Any hesitation in selected a flow post-merge will result in quicky losing sight and ultimately the fight. Taking angles when available will also make it easier to maintain sight post-merge. Fighters must remember to preserve the 500 ft safety bubble for training at all times.

<u>Bubble Relationships</u>:

It is important for the fighter to recognize the adversary's position relative to the fighter's bubble and vice versa.

1. Adversary outside the Fighter's bubble– When the adversary stops tracking aft on the fighter's canopy and starts moving forward.

2. Fighter outside the Adversary's bubble – Adversary's target aspect is decreasing.

Initiating Flows:

After controlling the merge, the fighter will set the flow. Below are general mechanics for the T-45C in initiating in plane and out-of-plane flows with respect to the adversary.

1. In-plane (<45°):

   a. 1C – For this flow, the fighter must have enough energy to counter a pure nose-high maneuver, but not so much airspeed such that when 1C flow is set, the fighter will flush out in front of the adversary. The fighter must utilize a combination of lift vector placement and an energy excursion to attempt to tighten its turn radius inside the adversary's. A good rule of thumb is to hit the merge with tactical vertical airspeed.

   b. 2C – It is best to be at corner airspeed. With an airspeed advantage at the merge, the fighter will be able to execute a max performance pull to instantaneously gain angles. If the adversary reverses 1C and the fighter assesses they will be defensive in the resulting in-plane 1C fight, the fighter must counter with an out-of-plane (OOPl) nose high maneuver.

2. Out-of-plane (>45°): Out-of-plane will result in one aircraft nose-high and the other nose-low or vice versa:

   a. OOPl 1C:

      i. Nose-low – This flow is optimum to force a low-to-high merge. Continue in the current direction of turn, utilizing a nose-low (approximately 45°) nibble of buffet pull to maintain energy on the jet, and to maximize turn rate while minimizing turn radius. The fighter must maintain enough energy to have the ability to pitch nose-high to meet the adversary at the next merge. Prior to 90° TA, the fighter should place the lift vector slightly below the adversary and pitch up into the fight. By placing the lift vector below the adversary, the fighter is attempting to make the merge more vertical. If excess energy exists, max perform to make the merge tight and steep.

      ii. Nose-high – The fighter should maintain a lift limit pull in order to tighten its turn circle inside that of the nose-low adversary (minimize turn radius). If the fighter assesses the adversary does not have enough energy to make the next merge, the fighter should put the lift vector in lag above the adversary to create exclusive-use turning room. If the fighter assess the adversary can make the next merge the fighter must get the nose back down and continue pulling to get the nose below the adversary in order to flatten out the merge. The fighter should strive to make the merge wide and flat.

b. OOPl 2C:

i. Nose-low – Initially, it is best to be at corner airspeed to gain angles after the initial merge. Max perform across the adversary's tail and wait for aft line of sight rates to stop and just start to move forward (occurs at the aft visual limit in the T-45C) before maneuvering out-of-plane. To maneuver out-of-plane, overbank 135 degrees and max perform targeting at least 45 degrees nose-low. In most cases with lateral merge geometry, maneuvering nose-low will be the best 2C flow (airspeed and altitude dependent). This presentation allows the fighter to trade altitude for turn performance and it achieves out of plane maneuvering. How nose-low the fighter commits are dependent upon airspeed and altitude available. Utilize a max performance pull targeting 500 ft above the deck. If above the rate band, the fighter should maintain a more level turn to bleed your airspeed. Approaching the rate band, the fighter should commit nose-low to maintain airspeed within the rate band, with a max performance pull As the adversary pitches down for the next merge, the fighter should adjust the lift vector to make the next merge happen and take angles. Expect this merge to be wide and shallow.

ii. Nose-high – 2C nose high vs. 2C nose-low is not advisable, as the T-45C will quicky lose energy nose-high and be unable to sustain airspeed in the rate band. Additionally, this sets the fighter up for unfavorable high-low merge geometry. To initiate this flow the fighter will need airspeed above the rate band, in excess of 370 KIAS. The fighter will need to max perform across the adversary's tail utilizing an oblique, nose-high pull trading airspeed for altitude. Once airspeed bleeds down to the rate band, the fighter should overbank to maintain your airspeed and pitch down to make the next merge.

701. UNIQUE MERGES

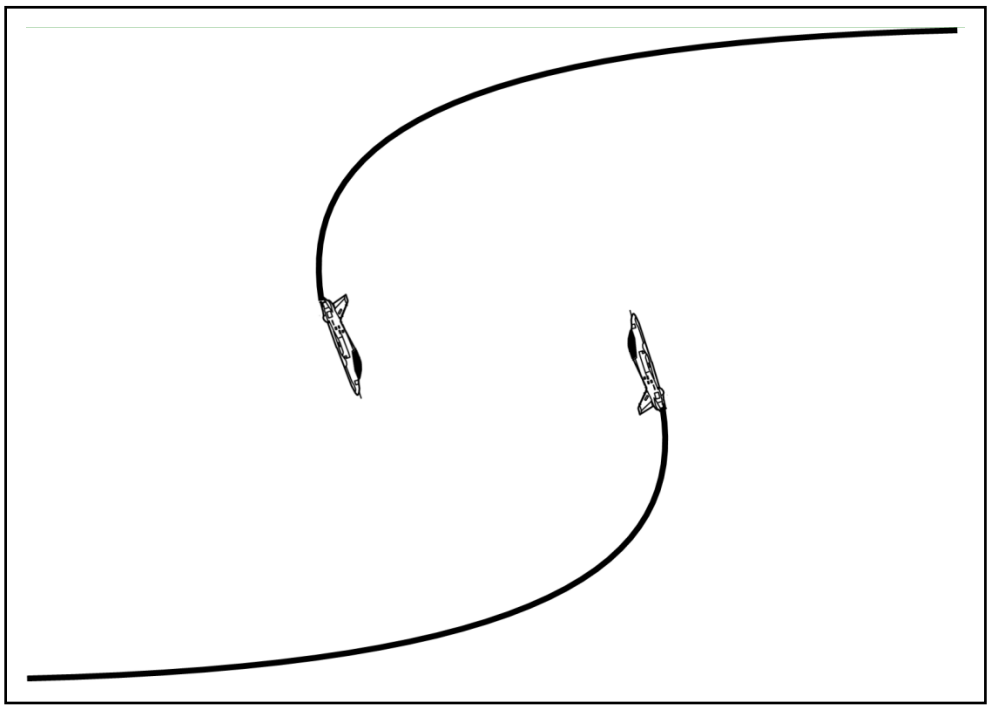
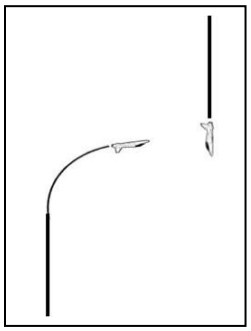


Figure 7-1 Unique Merges

Merges with a significant vertical component (> 45°):



Low to high – This is a highly desirable merge for the fighter. The fighter should attempt to make this merge as steep as possible by placing the lift vector underneath the adversary prior to pitching up to make the merge. The fighter must ensure they have tactical vertical airspeed at the most nose-low point and altitude available above the deck prior to steepening up the impending merge. Ideally, the low fighter should maneuver to stack the merge underneath the adversary, but some lateral turning rooming is acceptable provided the merge is sufficiently steep and tight. The fighter must execute an early turn ***prior*** to the merge while honoring the 500 ft bubble and sight of the adversary is maintained, (i.e., no blind lead turns). Prior to the merge, the fighter will roll to place adversary on one side of the canopy and execute a nibble of buffet pull to ideally have their nose on the horizon as the adversary passes through the horizon. This degree of early turn is difficult to accomplish in the T-45C due to its lack of nose authority and turn performance. But, the same result can be accomplished by early turning slightly in the direction that sets 1C flow. Maintain a nibble of pull (to prevent pitch buck) to keep the nose tracking back down to the adversary. In this moment, the force of gravity is hindering the adversary's turn performance while minimizing the turn radius of the fighter. Place the lift vector slightly in front of the adversary and employ with a LAR available. Post-shot, maneuver to lag and transition to OBFM. If unable to employ, prioritize maneuvering aft of the adversary's 3/9-line to gain the offensive position. If unable to get the nose back down, be

prepared to transition to a rolling scissors as a series of horizontal and vertical overshoots may occur. If the adversary succeeds in setting 2C flow, the fighter must expeditiously get the nose back down to gain airspeed within the rate band.

High to low – The fighter must strive to shallow out the merge as this geometry is highly undesirable. Prior to the merge, the fighter must place the lift vector, and subsequently the nose below the adversary to attempt to shallow out the merge and then slowly recover from the nose-low attitude prior to the merge. This merge must be as wide and shallow as possible. If the merge still has a substantial vertical component and the adversary executes an early turn, the fighter must counter by performing a lift limit pull and subsequently maneuver to set 2C flow. This will minimize the vertical overshoot. If the adversary extends in the vertical (unlikely in the T-45C), unload for vertical airspeed and attempt a pure nose-high maneuver. If able to significantly flatten the merge, turn across the adversary's tail and assess. If the merge is shallow and wide, and the adversary early turns away, capitalize on the BFM error by max performing 1C to gain the positional advantage. If the adversary maneuvers to set 2C flow, the fighter should have an energy advantage in the ensuing 2C fight.

Slow-speed merges:

A slow-speed merge occurs when neither aircraft has vertical airspeed. In the T-45C, this commonly occurs after both fighters maneuver pure nose high and meet at a subsequent lateral merge. It is preferable to arrive at this merge stacked slightly high to preserve a slight positional advantage. Pull to a 500 ft pass and continue to turn across the adversary's tail being sure to not descend (best possible turn performance in the T-45C in this regime of flight). Continue this turn until the adversary is outside of the bubble. For two similarly performing T-45Cs, bubble separation will occur simultaneously as the fighters reach their respective aft visual limits. The adversary's aft line of sight rate will slow, stop, pause, and very slowly begin to move forward from the fighter's perspective. Simultaneously, the fighter will see decrease in the adversary's target aspect (same sight picture but opposite perspective). Once the adversary exits the fighter's bubble, it is time to redefine. Unload, roll, and place the lift vector on the adversary redefining through pure nose low (with altitude available) while capturing a nibble of buffet pull (to avoid pitch buck). If the adversary simultaneously redefines, expect to meet in a lateral merge heading to the deck. If the adversary is late to redefine or is shallow, vary the lift vector slightly to lag to maneuver aft of the adversary's 3/9-line if possible or at the very least, maneuver to gain angles at the next lateral merge. If the adversary executes the redefinition early, continue to pull level across the horizon as the adversary has effectively executed an ICFPOS in the vertical. Wait until LOS rates accelerate and roll to place the lift vector on to slightly in front to max perform and take a shot.

702. GAME-PLAN DEVELOPMENT

Aircrew should enter every high-aspect fight with a game plan looking to exploit the weaknesses of the adversary while capitalizing on the strengths of their aircraft. A game plan should include the fighter's initial energy package, desired merge geometry, the follow-on flow, and reactions based off the adversary's initial maneuver. Finally, aircrew should strive to think one merge ahead to develop a proactive fighting stance, instead of simply reacting to the adversary.

703. BUTTERFLY SETUP

The objective of the Butterfly set is for each fighter to arrive at a neutral, co-altitude, 500-1000 ft merge with an unknown airspeed. Each aircrew is free to modulate the throttle to achieve the airspeed desired at the merge to best suit their preferred post-merge game plan. Aircrew will have to utilize BFM cues to assess the other aircraft's energy state and fight accordingly. In the T-45C, it is recommended to select MRT during the cut away and leave the engine staged at MRT for the duration of the engagement. This will aid in compressor stall mitigation and result in an airspeed at the merge that is acceptable for all follow-on game plans in the T-45C.

1. Both aircraft will climb to PADS altitude at 300 KIAS. When Lead calls "set 350," select MRT. Adjust the throttle to arrive at PADS altitude and airspeed. At this point, Lead will initiate "speed and angels."

2. Lead will then call to take a cut away. Both aircraft will take a 30° cut away from each other. During the cut away, each fighter is free to adjust airspeed for the desired merge game plan.

3. At 3.0 nm or visual limits (whichever occurs first), Lead will call turning in and call a pass, which the wingman will then echo. The fighters will turn into each other for a cooperative merge.

4. Both fighters will adjust their pulls to set the neutral pass.

5. Automatic "fight's on" at 3/9-line passage. Both fighters begin executing their HA game plan:

Adversary:	"Set 350"
	"Hammer 11 speed & angels left"
Fighter:	"Hammer 12 speed & angels right"
Adversary:	"Take a cut away"
	"Hammer 11 turning in"
	"Left-to-left/Right-to-right"
Fighter:	"Hammer 12 turning in"
	"Left-to-left/Right-to-right"

Butterfly lost sight game plan: If you go blind during the turn in, call it immediately. Lead will talk you to the merge. Listen to Lead's directions and execute them, essentially letting Lead fly your jet. Once sight is regained, call it. If both aircraft are blind, Lead will climb 1,000 ft and Wing will descend 1,000 ft with a KIO call from Lead.

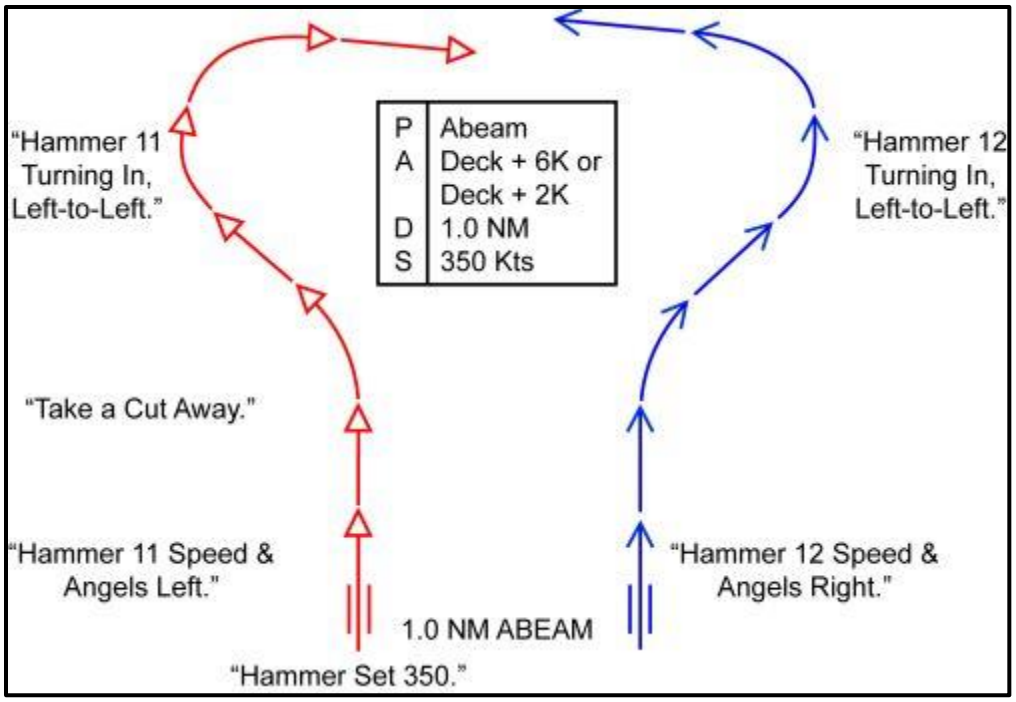


Figure 7-2 Butterfly Setup

704. ABEAM SETUP

For the Abeam set, the fight is on right away and is a neutral, known airspeed start. The initial abeam setup merge need no longer be co-altitude, allowing the fighters to add or subtract energy through the use of descending or climbing pre-merge. This allows each fighter to execute their respective BFM game plan. Each fighter should look to control that first merge to execute their HA game plan.

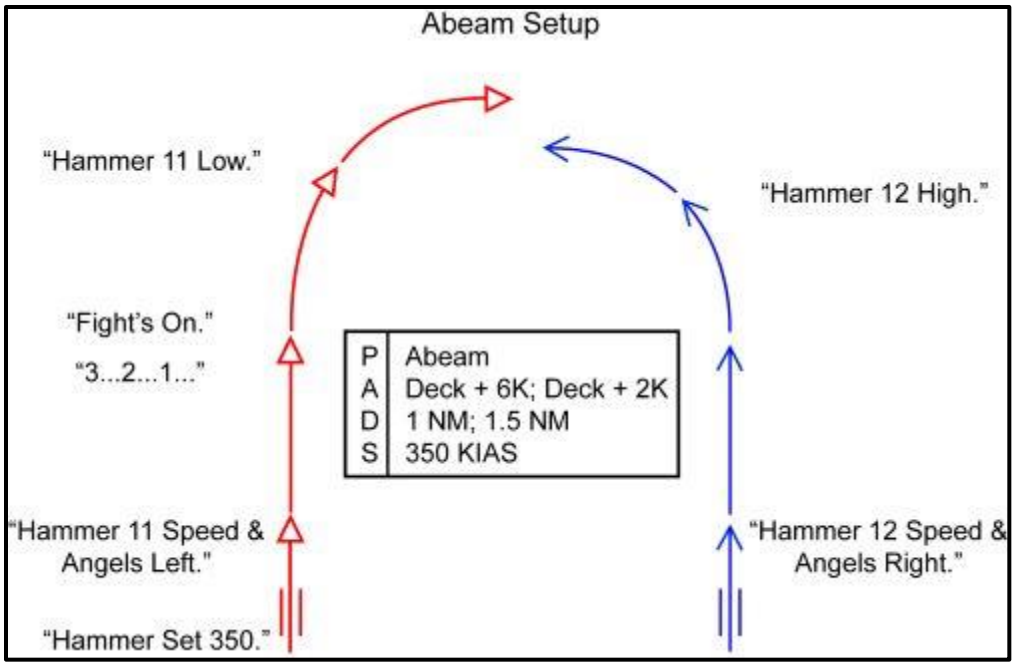


Figure 7-3 Abeam Entry

1. Both aircraft will climb to PADS altitude in combat spread at 300 KIAS. When Lead calls "set 350," select MRT. Adjust the throttle to arrive at PADS altitude and airspeed. At this point Lead will initiate "speed and angels."

2. Following this call, Lead will initiate the fight by calling "3, 2, 1 fight's on."

3. On the "3" both aircraft will select MRT.

4. At "fight's on," both aircraft are free to maneuver, ***and Lead will immediately call the pass for the first merge***.

5. Adversary: "Set 350"
"Hammer 11 speed and angels left"
Fighter: "Hammer 12 speed and angels right"
Adversary: "3, 2, 1 fight's on"
Adversary: "Hammer 11 High/Low"
Fighter: "Hammer 12 Low/High"

705. PREFERRED T-45C HIGH-ASPECT BFM GAME-PLANS

Butterfly (Deck +6): Max perform across the adversary's tail to take out the turning room, maintain sight, and wait for aft line of sight rates to slow down and just begin to translate forward at or near the aft visual limit. Assuming a neutral start and a well-executed max performance pull across the adversary's tail, this should occur as the adversary reaches the aft visual limit post-initial merge in the T-45C. Aft line of sight rates translating forward signify that the adversary is outside the fighter's bubble and the fighter should now maneuver out-of-plane. Unload, roll, and overbank 135 deg striving to achieve at least 45 deg nose low while re-applying a max performance rumble of buffet pull while padlocking the adversary and assessing their initial move. Airspeed should remain in the upper end of the rate band (300-330 KIAS) heading to the deck.

If the adversary maneuvers pure nose high, execute low-high merge mechanics.

If the adversary stays level, continue to maneuver out-of-plane nose low maintaining a lift limit pull targeting 1,000-2,000' above the deck. A quick inside-outside scan is critical to not busting the deck. Once at the lowest point, re-orient the lift vector slightly in front of the adversary and continue to max perform, cutting across the bottom of the adversary's turn circle, re-pierce the adversary's bubble, and employ a Fox-2 in the beam. Immediately maneuver to lag and maintain the offensive position. This is an uncountered out-of-plane maneuver.

If the adversary maneuvers 2-C nose low, continue to max perform to 500' above the deck. Once on the deck, the adversary will continue to move forward on the canopy. Continue to max perform to a 500' pass and capture airspeed once the pass is set. Assuming the fighter did not ease, encounter pitch buck, or execute a deck save, the fighter should arrive at a neutral lateral merge on the deck most likely below 300 KIAS. Due to airspeed now being below min vertical, level slow-speed merge mechanics apply. Post-merge, continue to capture until aft line of sight rates slow down and begin to translate forward and execute a nibble of buffet pull (energy excursion) to continue to pull the adversary forward on the canopy to take/deny angles. This will bleed airspeed but possibly gain angles at the next merge if the adversary does not recognize bubble separation and the need for an energy excursion. After one or more slow speed merges on the deck, the adversary may demonstrate a purposeful error by reversing 1-C nose high yielding lateral turning room to be taken by the fighter. Max perform with the lift vector up and aft of the adversary and kill!

Abeam (Deck + 6): Many potential game-plans exist for this scenario since the initial merge is non-cooperative, controllable by the fighter, and occurs in the center of the vertical airspace (assuming Deck to Deck +12/13). That being said, one particular T-45C game-plan has more possible positive outcomes than the others. This in no way invalidates other sound game plans. For example: If the low fighter has significant angles nose high underneath the adversary, it would be prudent for the low fighter to continue to capitalize on those angles by maneuvering pure nose high, cutting across the top of the adversary's turn circle, re-orienting the LV in front of the adversary, rapidly re-piercing the bubble, and employ weapons assuming the fighter can get the nose down before the adversary gets their nose up.

Preferred Game-Plan

If given the option to call "high" on the initial pass, the fighter should strive to control the merge by pulling harder to the inside of fight center than the adversary. The fighter will climb slightly to honor the pass called but also gain lateral turning room to the inside of the fight. For example, if starting on the left of the formation, the fighter should pull slightly oblique nose high to the inside of the merge to honor the "high" but also create some lateral separation with a left-to-left component. The key to success is to control this geometry and not let the adversary stack the merge low. After 3/9-line passage, maneuver oblique NL across the adversary's tail and assess their reaction.

If the adversary maneuvers pure nose high, execute low-high merge mechanics.

If the adversary maneuvers 2-C nose low, execute 2-C nose low mechanics as discussed above.

If the fighter does a poor job of controlling the merge and allows the adversary to stack the merge below, attempt to set 2-C flow but not at the expense of losing sight. If the adversary succeeds in gaining angles underneath and maneuvers pure nose high or sets oblique 1-C flow, look to match in the 1-C fight or maneuver nose low to set up for a low-high merge.

If the adversary is allowed to stack the merge low, there is a high likelihood (in the T-45C) that the high fighter will lose sight post-merge. By calling "high" and merging with a lateral component, the fighter maximizes their chances of maintaining sight after the adversary's initial move.

Butterfly (Deck + 2): Max perform level across the adversary's tail to take out the turning room and assess the adversary's first move. When presented with a neutral lateral merge, there is the potential to take angles post-merge if the adversary does not match. Also, maneuvering across the tail will ensure the fighter maintains sight and could set up for a very advantageous low-high merge. If the fighter were to max perform across the adversary's tail and maneuver pure nose high, there is a high likelihood that the fighter will lose sight momentarily and may experience difficulty in regaining sight if the adversary stayed level on the deck vice matching pure nose high. Also, it is common for the first look (left or right) over the shoulder to find adversary when maneuvering pure nose high is the wrong direction if an inaccurate assessment of whether the adversary crossed the fighter's extended 6 o'clock post-merge is made. This could result in the fighter going blind and the lift vector coming off the adversary prior to a slow-speed merge which will yield angles to the adversary.

If the adversary maneuvers pure nose high, continue to max perform underneath the adversary to steepen the impending low-high merge. The fighter may use the 2,000' above the deck to facilitate a steep merge. Execute low-high merge mechanics and kill! Depending on the altitude at which the low-high merge occurs, the fighter may not have enough altitude to commit as nose low as may be required to take a shot and save the deck. Prioritize maneuvering to gain the offensive position.

If the adversary maneuvers 2-C down to the deck. Execute 2-C nose low mechanics as discussed above.

7-12 HIGH ASPECT BFM

Abeam (Deck +2): If given the option to call "low" on the initial pass, the fighter should strive to control the initial merge by stacking the merge underneath the adversary with angles. At "fights on", the fighter needs to overbank and pull as required to descend below the adversary. Approximately half-way to the merge and with appropriate step down to make a safe pass, the fighter should preemptively pull the nose up to take angles in the vertical pre-merge. With significant angles underneath the adversary, the fighter should continue to capitalize on those angles taken and continue to maneuver pure nose high. Another advantage in taking angles underneath and not accepting a neutral low/high pass is that the adversary will remain further forward on the fighter's canopy post-merge and with angles and sound nose high mechanics, the fighter has a high likelihood of maintaining sight throughout the pure nose high maneuver.

If the adversary matches pure nose high, maintain lift vector on, honor the earth-stabilized pass, and make the merge. If the adversary's lift vector comes off or the adversary floats the pull to the vertical, look for opportunities to employ weapons. If the merge is lateral, strive to be stepped up (to gain exclusive-use turning room) and execute slow-speed merge mechanic as discussed previously in this chapter. If low, stack the merge underneath the adversary and reactively set 1-C flow. When considering the sum total of the energy packages of each fighter (potential plus kinetic energy), it benefits the low fighter to reactively set 1-C flow vice allowing a 2-C fight with a higher energy adversary. Match the adversary's turn direction and set 1-C flow, remain level, and assess the adversary's reaction. If the adversary commits nose low and flushes out in front, max perform and kill!

If the adversary maintains level on the deck, there is an opportunity to capitalize on an uncountered out-of-plane maneuver by maneuvering pure nose high. With significant angles taken underneath the adversary at the initial merge, there is an opportunity for the fighter to get the nose back down and on to the adversary prior to the adversary being able to get the nose back up. For the T-45C, recommend a nibble of buffet pull through pure nose high. Attempts at a max performance pull will likely result in pitch buck and the rapid loss of energy/airspeed near pure nose high potentially resulting in going ballistic or a departing controlled flight in the T-45C. Once through the most nose high point, re-orient the lift vector to slightly in front of the adversary and cut across the top of the adversary's turn circle. Max perform to quickly pierce the adversary's bubble and employ. Maneuver to lag and maintain the offensive position. Caution: If insufficient angles were attained beneath the adversary at the initial merge, the pure nose high maneuver was poorly executed, the lift vector not effectively updated, and the adversary's bubble not re-pierced, the adversary will likely be able to make the next merge, setting up the fighter for a disadvantageous high-low merge. Execute high-low merge mechanics as discussed previously in this chapter.

706. HIGH-ASPECT BFM CONCLUSION

The training command BFM stage is designed to give the student a solid foundation of concepts and definitions, as well as introduction on how to apply them in the air. While the student's next platform will be much more advanced in both performance and weapons systems, the lessons learned here will provide a fundamental basis for BFM engagements.

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APPENDIX A
GLOSSARY

A100. GLOSSARY

Admin Lead: The flight lead.

Adversary: Aircraft identified as an enemy, in CNATRA this will be the flight lead.

Angels: Altitude of aircraft in thousands of feet.

Angles Off Nose (AON): Angular difference between the attacker's twelve o'clock (longitudinal axis), and the defender's position. The attacker looking directly nose-on to the defender would be zero degrees AON.

Angles Off Tail (AOT): Angular difference between the defender's extended six o'clock (longitudinal axis), and the attacker's position. Directly behind the defender would be zero degrees AOT.

AOB: Angle of bank.

Arcing Turn (Arc): A turn executed at less than the optimum rate of turn.

Aspect: Angular description of an aircraft in reference to the fighter's current position.

Attack Window: A piece of sky located aft defender's bubble where, if an attacking aircraft max performs at the right time, they may have the opportunity to arrive in the defensive aircraft's CZ with RAC under control.

BFM: Basic fighter maneuvering (synonymous with air combat maneuvering for purposes of the training command).

Belly Check: Changing AOB on the aircraft to check areas masked from view by your own aircraft.

Beyond Visual Range (BVR): Situation where an intercept through radar or GCI identifies a group that is beyond the visual ACM arena.

Blind: Call from fighter meaning, "I do not see my lead/wingman/friendly."

Bubble: A representation of an aircraft's turn circle, in all three dimensions, if max performing at the aircraft's current energy state.

Check Left/Right (Degrees): To alter heading any number of degrees to the left or right.

Control Zone (CZ): A cone-like area, 2,000 ft to 4,000 ft behind a maneuvering aircraft, 20 degrees wide at the front to 40 degrees wide on the back side, centered on the aircraft's flight path where if an attacking aircraft arrives with angles and closure under control, the defensive aircraft can do nothing to deny the attacker the positional advantage.

Corner(ing) Airspeed: The airspeed at which the lift limit meets the load limit (410 KIAS @10,000 ft); this airspeed provides the best instantaneous turn rate, at the highest bleed rates.

Degrees to Go: The number of degrees of turn that an offensive aircraft has to turn in order to enter a suitable weapons envelope.

(The) Egg: A three-dimensional oval showing the effects of gravity on an aircraft maneuvering in all three planes.

Energy Package: The combination of the aircraft's altitude (potential energy) and airspeed (kinetic energy), establishing the aircraft's total energy.

Energy Rate Deck Transition: Trading available altitude for airspeed as a result of low fuselage alignment (neutral fight); reference the “ten percent rule.”

Engaged Fighter: In multi-plane engagements, the aircraft that is fighting an aggressive 1v1 against the bandit; the bandit’s lift vector placement will determine which fighter is engaged.

Engaging Turn: An efficient combination of turn rate and radius that maintains energy.

Engaging Turns: The type of turns used by a section to engage an enemy contact, i.e., Tac turns (both into and away), in-place turns, and cross turns.

Extension: A maneuver performed to achieve either range and/or angular separation to employ weapons or exit an engagement.

Feet Dry/Wet: Flying over land or water.

Flat Scissors: Defensive maneuver used to take advantage of an attacker's horizontal overshoot; also results from the flattening of rolling scissors, most likely due to deck proximity.

Flight Path: The imaginary arc that an aircraft scribes in the sky. The aircraft's velocity, G, and LV placement determine the geometry of the flight path. An aircraft that is straight and level has a straight flight path while one in a hard break turn has a very dynamic flight path. The smoke from an aircraft's engine is an excellent indication of an aircraft's flight path.

Flow (1C/2C): When two aircraft meet head-on, one of two types of flow is established in a turning fight after the merge occurs. If, after the merge, both aircraft turn across each other’s tail, the flow is said to be "2C" because each aircraft is still on its own distinct turn circle. Both aircraft will be turning the same direction, i.e., both in a left-hand turn; note that in 2C flow, the two aircraft are fighting nose-to-tail. If, at the merge, one aircraft reverses towards the other aircraft, the flow is now said to be "1C" because both aircraft are now on the same turn circle but

in opposite directions, i.e., one is turning left, the other is turning right. This is described as a nose-to-nose fight.

FOX-2: A call indicating the release of an IR (heat-seeking) guided missile made by the aircraft releasing the missile.

Hard Turn: Compromise between a maximum rate turn and energy conserving turn (17 units AOA), typically, the nibble of buffet.

High Yo-Yo: Offensive maneuver designed to hold or increase range by decreasing closure rate and opening nose-to-tail in low-to-medium angle off situations.

Intentions (in terms of ACM training rules): This term is used to ensure aircraft have de-conflicted flight paths when converging (e.g., "Lion 11, High/Low/Left-left/Right-right").

Joker: Fuel state above Bingo fuel which allows for two minutes at MRT, which would allow a successful bugout, normally transmitted to notify lead/wingman.

Knock-it-off: Call made to stop the fight or current maneuvers.

Lateral Separation: Lateral distance between two aircraft.

Lift Vector: The vector created through the production of lift, perpendicular to the wing-plane; the vector extends vertically, perpendicular to your waterline. If straight and level, it would point pure vertical. Picture yourself looking directly up and through the MDC on the canopy; used to define pursuit curves when in a different plane of motion from the adversary.

Line of Sight (LOS): An imaginary straight line from the fighter's eye to the adversary's aircraft.

Low Yo-Yo: An offensive maneuver designed to decrease range and angles by increasing closure rate, typically through the use of out-of-plane maneuvering.

Lufberry: Horizontal or slightly oblique, stalemate-type engagement where both aircraft are across the circle from each other, turning in the same direction at a low energy state, typically on the deck.

Merge: RADAR tracks have come within three nautical miles of each other; in the visual arena, used to describe two aircraft coming to a close-aboard pass.

No Joy: Call made meaning "I do not see the adversary."

Nose-to-Tail: Reference to the distance between the nose of an attacker and the tail of the defender. It is used synonymously with range.

1C Flow: An engagement between two aircraft that are turning nose-to-nose, through opposite AOB. This fight is referred to as a "radius fight" since the aircraft with the smallest turn radius is likely to achieve the advantage.

Overshoots: Flight path - occurs anytime the offensive aircraft flies through the defensive aircraft's flight path, at or aft of the defensive aircraft's 3/9-line.

3/9-Line - occurs anytime the attacker flies from aft of the defender's 3/9-line, to in front of the defender's 3/9-line (i.e., flushing out in front).

Out-of-Plane Maneuvering (OOP): Anytime your aircraft is maneuvering out of the plane of motion with respect to your opponent's plane of motion (> 45 degrees).

Padlocked: Call meaning I have a tally and cannot take my eyes off the adversary, or I will lose contact due to visibility/range, etc.

Plane of Motion (Plane of Turn): The flat plane of the turning circle. An aircraft's plane of motion is generally determined by assessing nose attitude and lift vector placement, or more simply put, the two-dimensional plane the aircraft is currently scribing.

Positional advantage: Describes one aircraft is aft of their opponent’s 3/9-line. A combination of angular advantage (i.e., less than 180-degrees of turn) to your opponent's flight path on the same heading, with 3/9-line advantage, and/or lateral turning room.

Positional Deck Transition: Trading available altitude for angles due to high fuselage alignment between defender and adversary (i.e., very defensive).

Post: The center of an aircraft's turn circle.

Pursuit Curves: Pursuit curves are based on the nose position when in the adversary's POM and the LV placement when not in the adversary's POM:

Lead – pointing in front of the adversary.

Pure – pointing at the adversary.

Lag – pointing behind the adversary.

Radius Fight: See 1C flow.

Range: Linear distance between two aircraft stated in nm or feet.

Rate Fight: See 2C flow

Rolling Scissors: Resulting fight with a series of horizontal and vertical overshoots with both aircraft in each other’s bubble, and neither fighter has energy to use the pure vertical.

State: Fuel remaining in thousands of pounds.

Shackle: Turn made to redress the section by crossing one member to other side, thus reassuming proper combat spread position.

Situational Awareness (SA): A person's perception of reality and how it reflects actual reality.

Skip It: Call made to indicate, "do not attack" or to "cease attack/intercept." In general, the offensive aircraft (shooter) owns in and down and the defensive aircraft (target) owns up and away from the merge.

Snap Shot Drill: Exercise designed to develop the skill of maneuvering into a snap guns solution, and firing on a moving adversary at high AOT.

Snap Guns: A non-tracking guns solution, targeting 60-90 degrees AOT, at a range of 1,000 ft to 1/2 mile.

Speed and Angels: A call made prior to an engagement to signify that the aircraft is in parameters to start the maneuver. Assumes for all engagements that all aircraft are in sight. This call will be made only when the aircraft is within the following parameters for the engagement:

+/- 100 ft altitude

+/- 0.1 miles

+/- 10 kts

+/- 10 degrees

Tac (Tactical) Lead: Member of the flight having the most SA and is responsible for directing the section's maneuvers. Note: this may not always be the admin lead.

Tac Wing: Member of the section not acting as the Tac Lead; assumes the role and responsibilities of the wingman.

Tally: Call meaning "a adversary visually sighted."

Three/Nine Line (3/9-line): A line drawn through the aircraft, perpendicular to the longitudinal axis through the aerodynamic center, which determines whether an opponent is in front of or behind that aircraft.

Ten Percent Rule: Used in any deck transition as an aid to avoid hitting the deck. Use 10% of altitude available above the deck to determine max nose-low depression
(i.e., 4000 ft above deck = 40 deg nose low, 3000 ft above deck = 30 deg nose low, etc.).

Track Crossing Rate: The speed at which the opposing aircraft appears to move across the fighter's canopy.

Turn Circle: The circle scribed by an aircraft's turn as it moves through the sky. The radius of this turn circle is constantly changing depending upon the "G" and velocity of the aircraft. We will generally discuss turn circle based on a generalized maximum performance turn as having a 3,000-ft radius, or 6,000-ft diameter.

Turning Room: Any separation that exists between two aircraft.

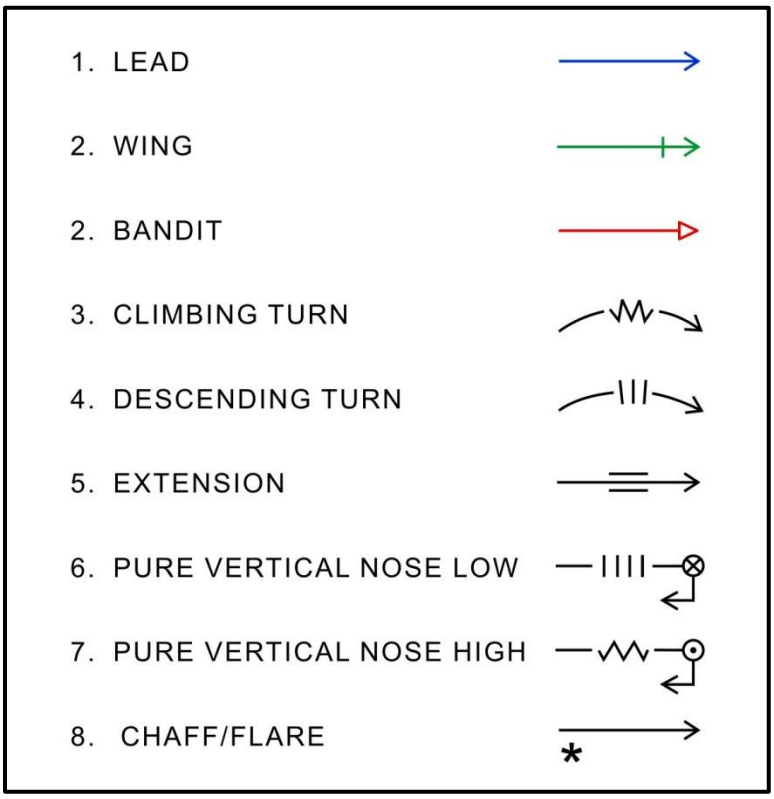
2C Flow: An engagement between two aircraft that are turning nose-to-tail through the same direction of turn. This is referred to as a rate fight because the aircraft with the faster rate of turn is likely to achieve the advantage.

Vertical 3/9-line: Line extending vertically 180 degrees out from the force of gravity, used to define the horizontal overshoot when in a different plane of motion from the adversary (e.g., when to execute the corkscrew maneuver in a rolling scissors).

Visual: Call meaning "wingman is in sight."

A101. SYMBOLOGY

You will see and use the following symbols in diagramming a BFM engagement.



APPENDIX B
ACRONYM GLOSSARY

1C: One-Circle

2C: Two-Circle

ACM: Aerial Combat Maneuvering

AGL: Above Ground Level

AGSM: Anti-G Straining Maneuver

AIM: Air Intercept Missile

ALOC: Almost Loss of Consciousness

AOA: Angle of Attack

AOB: Angle of Bank

AON: Angle off Nose

AOT: Angle off Tail

AW: Attack Window

AWE: Attack Window Entry

BFM: Basic Fighter Maneuvering

BVR: Beyond Visual Range

CNATRA: Chief of Naval Air Training

CS: Combat Spread

CV: Carrier

CZ: Control Zone

DBFM: Defensive Basic Fighter Maneuvering

DBT: Defensive Break Turn

DME: Distance Measuring Equipment

E-M: Energy-Maneuverability

EA: Electronic Attack

EBC: Eyeball Calibration

EGT: Exhaust Gas Temperature

EP: Emergency Procedures

ERDT: Energy Rate Deck Transition

FOV: Field of View

FQ: Forward Quarter

FRS: Fleet Replacement Squadron

FTI: Flight Training Instruction

G-LOC: G-induced Loss of Consciousness

G: Gravity

HABFM: High Aspect Basic Fighter Maneuvering

HUD: Heads-Up Display

IAW: In Accordance With

ICFPOS: In-Close Flight Path Overshoot

ICS: Intercom System

IMC: Instrument Meteorological Condition

IP: Instructor Pilot

IR: Infrared

KIAS: Knots Indicated Airspeed

KIO: Knock-It-Off

LAC: Lead Angle Computing

B-2 ACRONYM GLOSSARY

LAR: Launch Acceptability Region

LF: Load Factor

LOS: Line-of-Sight

LV: Lift Vector

MATC: Misaligned Turn Circles

MRT: Military Rated Thrust

MSL: Mean Sea Level

NATOPS: Naval Air Training and Operating Procedures Standardization

OBFM: Offensive Basic Fighter Maneuvering

OCF: Out-of-Control Flight

OOPh: Out-of-Phase

OOPl: Out-of-Plane

PADS: Position, Altitude, Distance, Speed

PDT: Positional Deck Transition

P_k: Probability of Kill

POM: Plane of Motion

RAC: Range, Angles, Closure

RG: Radial G

ROE: Rules of Engagement

ROT: Rule of Thumb

RPM: Revolutions per Minute

RQ: Rear Quarter

RTB: Return to Base

RTGS: Real Time Gun Sight

SA: Situational Awareness

SB: Speed-brake

SEM: Section Engaged Maneuvering

SN: Sensor Nose

SSD: Snapshot Drill

TA: Target Aspect

TACAN: Tactical Air Navigation System

TAC SOP: Tactical Standard Operating Procedures

TR: Turning Room

TRACOM: Training Command

VCR: Videocassette Recorder

VMC: Visual Meteorological Condition

WEZ: Weapons Engagement Zone

WSPAN: Wingspan